

Corrective biases in coordination economies with dispersed information

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Abstract

Can mistaken beliefs correct inefficient use of information? I study coordination economies with dispersed information in which private and social coordination incentives differ. When complete-information decisions are efficient, a common misperception of idiosyncratic precision can implement the constrained-efficient allocation. Overestimation provides the correction when private coordination incentives are too strong; underestimation when incentives are too weak. The optimal error depends on the payoffs but is invariant to the information structure. Correcting beliefs without correcting incentives can therefore lower welfare. In a two-period extension where actions generate information, efficiency requires different precision corrections across times. A constant error can still improve welfare but cannot implement the linear team optimum.

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1 Introduction

A common misperception of relative information precision can correct the inefficient use of dispersed information. Agents who respond to their signals do not internalize the full social cost of aggregate fluctuations and of cross-sectional dispersion, so privately optimal responses to information differ from socially efficient ones ([Angeletos and Pavan, 2007](#); [Angeletos and Lian, 2016](#)). When decentralized decisions are efficient under complete information, I show that there exists a unique belief error a belief that reproduces the constrained-efficient allocation. I characterize this error across a broad class of information environments and show how welfare changes with the size of the error.

Individuals often rely excessively on the quality of their private information. Biases of this kind, such as overconfidence or overprecision, have been extensively documented in the literature.¹ Executives report excessively narrow distributions for future stock returns, and their miscalibration is associated with more aggressive investment and financing decisions ([Ben-David et al., 2013](#)). Such mistakes can also be associated with worse individual outcomes, as the evidence on trading performance in experimental financial markets shows ([Biais et al., 2005](#)). Perceived precision matters for how agents combine different sources of news: [Broer and Kohlhas \(2024\)](#) show how over-reliance on private information can account for patterns of professional forecast revisions and departure from rational expectations. When decisions affect others, via either strategic complementarity or substitutability, however, the social consequences of these mistakes depend on the externalities they reinforce or offset. An error that worsens individual decisions can improve their collective outcome by compensating for a discrepancy between private and social incentives. This makes the interaction between perceived precision and coordination incentives central to evaluating such errors.

In a linear-quadratic Gaussian economy within the framework of [Angeletos and Lian \(2016\)](#), allowing for heterogeneous fundamentals and persistent information, I show that a common mistake about information can be welfare improving. Payoffs depend on individual and aggregate actions and on their cross-sectional dispersion. Agents share a proportional misperception of idiosyncratic precision relative to common precision and otherwise update consistently under their perceived model. I consider economies in which decentralized decisions are efficient under complete information, so the remaining inefficiency comes only from dispersed information. Following [Angeletos and Pavan \(2007\)](#); [Angeletos and Lian \(2016\)](#), the constrained-efficient benchmark maximizes expected welfare subject to each agent using only her own available information. It is the solution to a team problem: decision rules are chosen cooperatively before information is realized, but information cannot be pooled across agents ([Marschak and Radner, 1972](#)). I show that an appropriate common precision error implements this allocation without changing either payoffs or the information agents receive.

¹Overly optimistic forecasts ([Malmendier and Tate, 2005](#)), distortions in household saving behavior ([Pfäuti et al., 2024](#)), and excessive market entry ([Camerer and Lovo, 1999](#)), among others.

The direction of correction follows from the discrepancy between private and social coordination incentives. When agents' incentive to follow the crowd is too strong, overestimating idiosyncratic precision offsets the resulting emphasis on predicting others' actions. When society values coordination more strongly than individuals do, instead, underestimating that same precision provides the necessary correction. Through the equilibrium representation of [Huo and Pedroni \(2020\)](#), I show that a common adjustment of perceived precision changes information use in exactly the way required to offset these externalities.

Welfare gains do not require the precision error to match this correction exactly. Any error that moves information use toward the efficient allocation improves welfare relative to accurate beliefs, while an error in the opposite direction lowers it. Errors that overcorrect can also remain beneficial: although they achieve less than the maximum attainable welfare, a range of them still outperforms the rational-expectations allocation. Thus, the welfare assessment of a belief error depends on both its direction and its size relative to the underlying incentive distortion. The analysis identifies the best degree of misperception and characterizes which departures from it preserve a welfare gain.

A precision error can thus have the same effect on information use as correcting coordination incentives, the object of policy in [Angeletos and Pavan \(2009\)](#). This equivalence matters when evaluating efforts to correct mistaken beliefs. Eliminating a bias that partly offsets inefficient coordination restores accurate beliefs while leaving the externality intact, and therefore lowers welfare. Eliminating a bias larger than the corrective value can instead raise or lower welfare. Although payoffs determine the corrective value, the information environment determines whether a larger error remains preferable to accurate beliefs. Consequently, even with identical incentives and the same existing bias, a change in signal precision can reverse the welfare case for eliminating that bias.

When actions generate information, the correction must also account for what others subsequently learn ([Vives, 1997](#); [Bernardo and Welch, 2001](#)). I examine this additional social return in a two-period economy in which agents observe a noisy signal of earlier aggregate activity. A stronger early response to private information makes that observation more informative. Reproducing the team optimum subject to linear first-period rules then requires a stronger precision correction at the first time period than at the second. No constant error can deliver both corrections. Nevertheless, a permanent error can improve welfare: when private and social coordination incentives coincide in the static problem, sufficiently mild permanent overconfidence raises total welfare. The extension identifies a limit to exact correction and shows why beneficial bias can survive that limit.

Related literature. The efficient benchmark builds on the distinction between private and social coordination in [Angeletos and Pavan \(2007\)](#); [Angeletos and Lian \(2016\)](#). Their analysis explains why rational agents can use information inefficiently even when decentralized decisions would be efficient under complete information. The representation of [Huo and Pedroni \(2020\)](#) provides a single-judge solution of coordination economies with dispersed information; I use this solution to identify the mistaken beliefs that implement the efficient allocation. The

welfare analysis also establishes when errors that fall short of, or exceed, the efficient correction still improve welfare relative to accurate beliefs.

Can a belief error correct inefficient information use without introducing another distortion? [Bizzarri and d'Arienzo \(2024\)](#) show that overreaction can improve welfare by encouraging trades that reveal private information through prices. In their market, the diagnostic bias that makes the response to private information efficient differs from the bias that makes the response to price information efficient, so correcting one response leaves the other distorted. In the coordination economies studied here, with exogenous information and efficient complete-information decisions, equilibrium and efficiency forecast the same target and differ in one scalar, so one relative-precision error corrects the entire decision rule. The required error depends only on the gap between private and social coordination incentives, and stays the same across information structures with different signal precisions, correlations, and persistence. The two-period extension connects this result back to the informational role of actions: when current decisions affect subsequent learning, the efficient corrections differ across periods, so a constant precision error can improve welfare without implementing the linear team optimum.

This paper also shows that incorrect beliefs can achieve the same social outcomes as direct policy interventions. [Angeletos and Pavan \(2009\)](#) characterize policies that correct incentives under dispersed information, while [Basak and Zhao \(2026\)](#) study optimal Pigouvian taxation when dispersed beliefs are also distorted. Such policies can address the externality alongside the decisions induced by mistaken beliefs. Correcting beliefs alone leaves the externality in place. As emphasized by [Mullainathan et al. \(2012\)](#), that distinction matters for evaluating behavioral interventions. Here, the exact correction and the welfare ordering identify when restoring accurate beliefs would remove a beneficial offset to inefficient incentives. Beliefs are not necessarily an instrument available to a policymaker; the question is whether correcting an existing error improves decisions under the incentives faced by agents.

Section 2 describes the economy and the efficient benchmark. Section 3 characterizes corrective precision errors and their welfare effects. Section 4 studies correction when actions generate later information. Section 5 concludes.

2 The economy

This section establishes how private and social incentives affect the use of dispersed information. I describe an economy in which payoffs depend on individual actions, aggregate activity, and the dispersion of actions across agents. I then compare equilibrium decisions with those chosen by a planner who faces the same information constraints. The comparison separates inefficiency that would remain under complete information from inefficiency in how agents use their signals. When private and social complete-information decisions coincide, equilibrium and efficiency differ only in the coordination motive governing information use. Expressing both allocations as conditional expectations allows me to study the role played by the corrective precision bias in the next section.

2.1 Payoffs and information

A continuum of agents $i \in [0, 1]$ act in periods $t = 0, 1, 2, \dots$. In each period agent i chooses a scalar action $k_{it} \in \mathbb{R}$. The aggregate action is the cross-sectional mean $K_t = \int_0^1 k_{jt} dj$, and $\theta_{it} \in \mathbb{R}$ is an exogenous variable that is payoff-relevant to agent i , her fundamental, with cross-sectional mean $\bar{\theta}_t = \int_0^1 \theta_{jt} dj$. The period payoff is $U(k_{it}, K_t, \sigma_t, \theta_{it})$, where σ_t measures the cross-sectional dispersion of actions, and lifetime payoff is the discounted sum of period payoffs.

Assumption 1 (Payoffs). $U(k, K, \sigma, \theta) = U^0(k, K, \theta) + \frac{1}{2}U_{\sigma\sigma}\sigma^2$, where U^0 is quadratic, $U_{kk} < 0$, $U_{k\theta} \neq 0$, $|U_{kK}/U_{kk}| < 1$,

$$W_{KK} := U_{kk} + 2U_{kK} + U_{KK} < 0, \quad W_{\sigma\sigma} := U_{kk} + U_{\sigma\sigma} < 0,$$

and σ_t^2 is the cross-sectional variance of actions net of their response to own fundamentals, $\sigma_t^2 := \int_0^1 (\tilde{k}_{jt} - \int_0^1 \tilde{k}_{lt} dl)^2 dj$ with $\tilde{k}_{jt} := k_{jt} - \kappa_1 \theta_{jt}$ and $\kappa_1 := -U_{k\theta}/U_{kk}$. Payoffs are additively separable across periods.

A common shift in actions moves k_{it} and K_t together and has welfare curvature W_{KK} ; a spread that leaves the mean fixed has curvature $W_{\sigma\sigma}$, and the ratio of the two decides how a planner trades off information that agents share against information that differs across them. Dispersion is net of the response to the own fundamental, which keeps that response efficient. With a common fundamental, σ_t is the raw dispersion of actions. The slope of an agent's best response to aggregate activity,

$$\alpha := -\frac{U_{kK}}{U_{kk}} \in (-1, 1), \quad (1)$$

measures how much she wants her action to move with the actions of others (strategic complementarity), holding her own fundamental fixed. At $\alpha = 0$ the actions of others are irrelevant to her, and below zero she wants to move against them (strategic substitutability). I call α the private coordination motive. Under complete information the dispersion term plays no role, because no single agent moves σ_t , and the unique equilibrium action is linear in the two fundamentals:

$$\kappa(\theta_{it}, \bar{\theta}_t) = \kappa_0 + \kappa_1 \theta_{it} + \kappa_2 \bar{\theta}_t, \quad \frac{\kappa_2}{\kappa_1 + \kappa_2} = \alpha. \quad (2)$$

Here $U_k(0)$ denotes the derivative of U^0 in k at the origin, $U_K(0)$ the one in K , and $\kappa_0 = -U_k(0)/(U_{kk} + U_{kK})$; the coefficients follow from the first-order condition $U_k(0) + U_{kk}k_{it} + U_{kK}K_t + U_{k\theta}\theta_{it} = 0$ and its cross-sectional average. I call $\kappa(\theta_{it}, \bar{\theta}_t)$ the complete-information action. The identity on the right says that the share of its response running through the aggregate fundamental equals α .

Assumption 2 (Information). Agent i observes $x_{it} = M_t(L)\varepsilon_{it}$, where $M_t(L)$ is a matrix polynomial in the lag operator with square-summable coefficients, possibly time-varying, and independent of agents' actions. The shocks $\varepsilon_{it} = (\eta_t, v_{it})$ are Gaussian, independent over time and of each other, with covariance $\text{diag}(\Sigma_\eta, \Sigma_v)$, Σ_v positive definite; η_t is common and v_{it} is idiosyncratic, independent across agents, with $\int v_{it} di = 0$. The fundamental is $\theta_{it} = \psi_t(L)'\varepsilon_{it}$ with square-summable ψ_t , so that $\bar{\theta}_t$ depends on the history of η alone.

Assumption 2 covers any finite number of private and public signals, persistent fundamentals and noise, and heterogeneous fundamentals; what matters below is the split between idiosyncratic and common shocks, and a private signal whose noise is partly correlated across agents loads on both. Cross-sectional averages obey the exact law of large numbers: for any function of the common shocks and the own idiosyncratic shocks with finite second moment, the average across agents equals the conditional expectation given the common shocks, under the true distribution and under every perceived distribution introduced below. Information is exogenous: agents observe only x_{it} and no aggregate actions.²

Agent i chooses k_{it} to maximize $\mathbb{E}_{it}[U(k_{it}, K_t, \sigma_t, \theta_{it})]$, taking K_t and σ_t as given, where $\mathbb{E}_{it}[\cdot]$ conditions on her signal history and on knowledge of the signal process and of the strategies of others. Her first-order condition is linear in conditional means, and dividing it by $-U_{kk}$ gives

$$k_{it} = \kappa_0(1 - \alpha) + \kappa_1 \mathbb{E}_{it}[\theta_{it}] + \alpha \mathbb{E}_{it}[K_t]. \quad (3)$$

Because K_t averages the actions of others, forecasting it means forecasting their forecasts, and α sets how much these higher-order forecasts matter. An equilibrium is a strategy that satisfies (3) almost surely, with K_t the mean the strategy generates; Appendix A.1 states the definition. Under Assumptions 1 and 2 the equilibrium exists, is unique among symmetric strategies, and is linear in the signal history (Huo and Pedroni, 2020, Propositions 1 and 3).

The class contains the beauty contest of Morris and Shin (2002), a forecast-aggregation economy, an economy with a social cost of dispersion, and an efficient competitive economy; Appendix B describes these examples, along with an investment-spillover economy that fails the efficiency condition imposed below.

2.2 Equilibrium and efficient information use

For $s > 0$, let $\mathbb{E}_{it}^{(s)}[\cdot]$ denote the conditional expectation given agent i 's signal history computed with idiosyncratic covariance Σ_v/s , all other primitives as in Assumption 2. At $s = 1$, $\mathbb{E}_{it}^{(1)} = \mathbb{E}_{it}$. With an infinite signal history, $\mathbb{E}_{it}^{(s)}[\cdot]$ is a limit of finite linear combinations of signals, and its value on the actual signals is the same limit computed under the true distribution (Appendix A.7).

Proposition 1 (Equilibrium information use). *Under Assumptions 1 and 2, the unique equilibrium is*

$$k_{it} = \mathbb{E}_{it}^{(1-\alpha)}[\kappa(\theta_{it}, \bar{\theta}_t)]. \quad (4)$$

Proposition 1 is the single-judge solution of Huo and Pedroni (2020, Theorem 1) for a com-

²The restriction holds the information available for decisions fixed when comparing equilibria and policies. It can describe decisions made before aggregate outcomes become observable, but it excludes learning from past aggregate activity. An observed aggregate generally changes its informational content when decision rules change, so the planner must account for the effect of current actions on future information. Section 4 allows this feedback in two periods. A longer, possibly infinite, horizon would require tracking this feedback through subsequent decisions and observations.

mon fundamental and an equivalent form of their Proposition 3 for heterogeneous fundamentals. Equilibrium forecasts the complete-information action as if idiosyncratic precision were $1 - \alpha$ times its value, since under strategic uncertainty private information gets discounted exactly by the degree of complementarity. With the target held fixed, a change in α therefore acts on the rule exactly as a proportional change in idiosyncratic precision does.

Example 1 (Two signals). Let $\theta_{it} = \theta_t$ be common, $\theta_t \sim N(0, \tau_\theta^{-1})$, and let each agent observe a public signal $z_t = \theta_t + \zeta_t$ and a private signal $x_{it} = \theta_t + v_{it}$, with noise precisions τ_z and τ_v ; normalize $\kappa(\theta, \theta) = \theta$. Then (4) reads

$$k_{it} = \frac{\tau_z z_t + (1 - \alpha)\tau_v x_{it}}{\tau_\theta + \tau_z + (1 - \alpha)\tau_v},$$

the Bayesian forecast of θ_t with the private precision deflated to $(1 - \alpha)\tau_v$. The rule is the equilibrium of Morris and Shin (2002) and of Proposition 1 in Angeletos and Pavan (2007), whose weight on public information, $\gamma = \delta + \alpha\delta(1 - \delta)/(1 - \alpha(1 - \delta))$ in their notation, simplifies to $\tilde{\tau}_z/(\tilde{\tau}_z + (1 - \alpha)\tau_v)$ with $\tilde{\tau}_z := \tau_\theta + \tau_z$ once the prior is absorbed into the public signal.

The complete-information optimum is equal to the action a planner would choose if $(\theta_{it}, \bar{\theta}_t)$ were common knowledge, and is a linear function of the fundamentals (Lemma A.3 in Appendix A),

$$\kappa^*(\theta_{it}, \bar{\theta}_t) = \kappa_0^* + \kappa_1^* \theta_{it} + \kappa_2^* \bar{\theta}_t, \quad (5)$$

with $\kappa_1^* = \kappa_1$ because dispersion is measured net of $\kappa_1 \theta_{it}$: the response to the own fundamental does not enter σ_t , so the planner has no reason to change it (with raw dispersion, $\kappa_1^* = -U_{k\theta}/W_{\sigma\sigma} \neq \kappa_1$ whenever $U_{\sigma\sigma} \neq 0$). No agent knows $\bar{\theta}_t$, so the benchmark I use is constrained efficiency in the sense of Angeletos and Pavan (2007, Definition 3) and Angeletos and Lian (2016, Definition 16), a team problem as in Marschak and Radner (1972): the planner chooses, for each period, one measurable rule g_t applied symmetrically to all agents, to solve

$$\max_{\{g_t\}} \mathbb{E} \left[\sum_t \beta^t U(k_{it}, K_t, \sigma_t, \theta_{it}) \right] \quad \text{subject to} \quad k_{it} = g_t(x_{it}, x_{it-1}, \dots), \quad (6)$$

where $\beta \in (0, 1)$ is the discount factor, the expectation is ex ante over all shocks and agents, and K_t and σ_t are the mean and dispersion generated by g_t . With separable payoffs and exogenous information, (6) is the static problem of Angeletos and Lian (2016) solved period by period, with a unique solution (Lemma A.3(ii); Angeletos and Lian, 2016, Section 9.2) that I denote k_{it}^* and call the team optimum.

Lemma A.3 shows that the planner's first-order condition has the fixed-point form of the agents' problem with a social coordination motive in place of the private one, $k_{it}^* = \mathbb{E}_{it}[\kappa^*(\theta_{it}, \bar{\theta}_t)] + \alpha^* \mathbb{E}_{it}[K_t^* - \kappa^*(\bar{\theta}_t, \bar{\theta}_t)]$, where $\kappa^*(\bar{\theta}_t, \bar{\theta}_t)$ is the complete-information optimum averaged across agents. The social slope is

$$\alpha^* := 1 - \frac{W_{KK}}{W_{\sigma\sigma}} < 1. \quad (7)$$

I call α^* the social coordination motive. It rises as the social cost of aggregate volatility falls

relative to that of dispersion. A more concave social return to aggregate activity, U_{KK} more negative, lowers it, while a level externality U_K does not move it; a social cost of dispersion raises it.

Proposition 2 (Efficient information use). *Under Assumptions 1 and 2, the constrained-efficient allocation is*

$$k_{it}^* = \mathbb{E}_{it}^{(1-\alpha^*)} [\kappa^*(\theta_{it}, \bar{\theta}_t)]. \quad (8)$$

Equilibrium and efficiency can differ in their targets, $\kappa \neq \kappa^*$, or in their coordination motives, $\alpha \neq \alpha^*$. Angeletos and Pavan (2007, Proposition 4) show, in the two-signal economy of Example 1, that equilibrium is efficient if and only if neither difference is present.³

Assumption 3 (Efficiency under complete information). $\kappa = \kappa^*$.

Under Assumption 3 the only inefficiency, if any, depends on how agents use dispersed information. Equilibrium and team optimum then forecast the same object under the same signals but differ in one scalar; equilibrium scales idiosyncratic precision by $1 - \alpha$, the private value of coordination, and the planner scales it by $1 - \alpha^*$, the social value.

3 Corrective biases

A shared precision error can restore efficiency under dispersed information, provided that decentralized decisions would be efficient under complete information. The required error depends on the gap between private and social coordination motives. Errors that only partly close this gap also raise welfare, with some overshooting still being beneficial. I first establish the optimal precision error, then characterize the welfare effects of other precision factors and illustrate how the direction of correction differs across payoffs.

3.1 Relative-precision errors

All agents assign covariance Σ_v/b to the idiosyncratic shocks when their true covariance is Σ_v , for a common factor $b > 0$. They update by Bayes' rule under this misspecified model, with the same beliefs for fundamentals and for aggregate actions, and know everything else, including the strategies of others. I call b the precision factor. It scales the whole idiosyncratic block, so the part of a private signal's noise that all agents share keeps its true covariance.

Definition (Distorted equilibrium). *A b -distorted equilibrium is a Bayesian-Nash equilibrium of the game in which the common prior assigns covariance Σ_v/b to the idiosyncratic shocks: a strategy that satisfies (3) for every agent and almost every signal history when all conditional expectations, including the cross-sectional mean K_t induced by the strategies, are computed under the perceived covariance Σ_v/b .*

³ $\alpha = \alpha^*$ iff $U_{kK} + U_{KK} = (1 - \alpha)U_{\sigma\sigma}$, and $\kappa = \kappa^*$ iff $U_K(0)(U_{kk} + U_{kK}) = U_K(0)(U_{kK} + U_{KK})$ and $U_{K\theta}(U_{kk} + U_{kK}) = U_{K\theta}(U_{kK} + U_{KK})$, which equate κ_0 and $\kappa_1 + \kappa_2$ from (2) with κ_0^* and $\kappa_1^* + \kappa_2^*$ from Lemma A.3.

The precision error is fixed and common knowledge. Every agent knows that others share it and anticipates their strategies under the same perceived model. Agents update beliefs about fundamentals, but never revise b in response to evidence about signal quality.⁴ The perceived signal process satisfies Assumption 2, so the distorted equilibrium exists, is unique, and is linear by the same argument as under the true signals. The b -distorted allocation is this strategy applied to the actual signals. The strategy is linear, so the cross-sectional mean it generates is the same random variable under the perceived and the true model (Appendix A.7).

Proposition 3 (Distorted information use). *Under Assumptions 1 and 2, the unique b -distorted equilibrium is*

$$k_{it}^b = \mathbb{E}_{it}^{((1-\alpha)b)} [\kappa(\theta_{it}, \bar{\theta}_t)]. \quad (9)$$

The error compounds with the equilibrium discount. Write $s = (1 - \alpha)b$, which I call the effective scale, and $k_{it}^{(s)} = \mathbb{E}_{it}^{(s)} [\kappa(\theta_{it}, \bar{\theta}_t)]$. The rational equilibrium is $k^{(1-\alpha)}$, and under Assumption 3 the team optimum is $k^{(1-\alpha^*)}$.

Errors in either covariance block act the same way. Suppose agents assign covariances $c_\eta \Sigma_\eta$ and $c_v \Sigma_v$ to common and idiosyncratic shocks, $c_\eta, c_v > 0$, with means and loadings unchanged.

Corollary 1 (Relative covariance distortions). *Under Assumptions 1 and 2, these beliefs generate the same equilibrium allocation as Proposition 3 with $b = c_\eta / c_v$. In particular, a common proportional error, $c_\eta = c_v$, leaves the rational-expectations allocation unchanged.*

Only relative precision affects actions within this class of errors. A common rescaling of uncertainty leaves conditional means unchanged, whereas a change in the ratio of common to idiosyncratic uncertainty changes information use. These are errors about covariances followed by Bayesian updating, the overconfidence case in the distinction drawn by Bizzarri and d'Arienzo (2024). They do not encompass diagnostic overreaction or arbitrary errors about individual signals. Each resulting rule $k^{(s)}$ is the team optimum of a fictitious economy with the same complete-information optimum and social coordination motive $1 - s$ (Lemma A.5).

3.2 Exact correction

Theorem 1 (Correction by a uniform precision bias). *Let Assumptions 1–3 hold and define*

$$b^* := \frac{1 - \alpha^*}{1 - \alpha} > 0. \quad (10)$$

- (i) (Sufficiency.) *The b^* -distorted allocation coincides with the constrained-efficient allocation for every agent, almost surely, and the distorted aggregate action coincides with the efficient aggregate action.*

⁴Broer and Kohlhas (2024) document forecasters believing their private information to be more precise than it actually is, without ever correcting the perceived signal precisions.

- (ii) (Uniqueness for a given signal process.) b^* is the only $b > 0$ with this property if and only if the efficient rule uses idiosyncratic information. With a common fundamental, $\theta_{it} = \bar{\theta}_t$, this means that k_{it}^* loads on some idiosyncratic shock; with heterogeneous fundamentals it means that k_{it}^* uses idiosyncratic information beyond forecasting the agent's own idiosyncratic fundamental, in the sense made precise in the proof. If the condition fails, every $b > 0$ implements the efficient allocation.
- (iii) (Uniqueness across signal processes.) b^* is the only $b > 0$ that implements the constrained-efficient allocation for every fundamental and signal process satisfying Assumption 2, the loadings ψ_t included. If Assumption 3 is dropped and $\kappa \neq \kappa^*$, no $b > 0$ does so.

Part (i) sets the effective scale $(1 - \alpha)b$ equal to the planner's $1 - \alpha^*$. With equal targets, the precision bias matches the necessary optimal weight exactly. Part (ii) identifies the environments in which relative precision does not move the allocation, such as a common fundamental observed only through public signals. Part (iii) says that no factor other than b^* works for every signal process, and that with $\kappa \neq \kappa^*$ none does, because under complete information every b delivers κ .

At b^* no signal becomes more informative and no incentive changes. Each agent is wrong about the precision of her signals and maximizes under the original payoffs, yet the allocation is the planner's. Angeletos and Pavan (2007, Proposition 3) reach the team optimum by changing the coordination slope to α^* in a fictitious game, and Angeletos and Pavan (2009) by taxes contingent on aggregates. Here the belief absorbs the gap and does the same job. Part (i) only requires to write the fictitious game as a single-judge forecast in the manner of Huo and Pedroni (2020).

Precision errors can substitute for an incentive correction only when the inefficiency concerns information use. Basak and Zhao (2026) instead choose taxes to correct a negative aggregate externality when agents overestimate the precision of public and private news. Their tax adjusts incentives given distorted beliefs; here I hold incentives fixed and ask which relative-precision error reproduces efficient decisions. The distinction matters under complete information: a precision error then leaves actions unchanged and cannot correct $\kappa \neq \kappa^*$. Theorem 1(iii) states this limit for the bias class studied here.

3.3 Welfare away from the corrective bias

Welfare gains are possible also for biases away from b^* . Welfare is ex ante expected payoff, $\mathcal{W}(k) := \mathbb{E}[U(k_{it}, K_t, \sigma_t, \theta_{it})]$, evaluated at a strategy k under the true information. Let $e_{it} := k_{it} - \kappa^*(\theta_{it}, \bar{\theta}_t)$ be the gap between an agent's action and her complete-information optimum and $\bar{e}_t := \int_0^1 e_{jt} dj$ its cross-sectional mean. Because U is quadratic, the expansion around (5) is exact and its first-order terms vanish (Lemma A.4):

$$\mathcal{W}(k) = \mathcal{W}^{\text{ci}} - \frac{1}{2}(-W_{\sigma\sigma}) \underbrace{\mathbb{E}[(e_{it} - \bar{e}_t)^2]}_{\text{Disp}(k)} - \frac{1}{2}(-W_{KK}) \underbrace{\mathbb{E}[\bar{e}_t^2]}_{\text{Vol}(k)}, \quad (11)$$

where $\mathcal{W}^{\text{ci}}$ is first-best welfare, i.e. the welfare at κ^* , which equals complete-information equilibrium welfare under Assumption 3. Equation (11) extends condition (9) of Angeletos and Pavan (2007) to heterogeneous fundamentals; its two moments are what they call dispersion and volatility, and $\text{Disp} = \mathbb{E}[\sigma_t^2]$ because $\kappa_1^* = \kappa_1$. I call the ratio $-W_{KK}/(-W_{\sigma\sigma}) = 1 - \alpha^*$ the social price of volatility.

Along the family $k^{(s)}$, write $D(s) := \text{Disp}(k^{(s)})$ and $V(s) := \text{Vol}(k^{(s)})$. The loss relative to the team optimum is

$$\begin{aligned} L(s) &:= \mathcal{W}(k^*) - \mathcal{W}(k^{(s)}) \\ &= \frac{1}{2}(-W_{\sigma\sigma}) \left\{ [D(s) + s^* V(s)] - [D(s^*) + s^* V(s^*)] \right\}, \quad s^* := 1 - \alpha^*. \end{aligned} \quad (12)$$

Lemma A.5 shows that $k^{(s)}$ uniquely minimizes $\text{Disp}(k) + s \text{Vol}(k)$ over feasible strategies, so the family traces a frontier between dispersion and volatility on which s is the price of volatility. Welfare evaluates every point at the social price s^* .

Proposition 4 (Welfare along the effective-information margin). *Let Assumptions 1–3 hold.*

- (i) $V(s)$ is nonincreasing and $D(s)$ is nondecreasing in s .
- (ii) $L(s)$ is nonincreasing on $(0, s^*]$ and nondecreasing on $[s^*, \infty)$, with $L(s^*) = 0$.
- (iii) If the efficient allocation uses idiosyncratic information in the sense of Theorem 1(ii), the monotonicity in (i)–(ii) is strict and $L(s) > 0$ for every $s \neq s^*$. Otherwise $L \equiv 0$.

Raising s moves the minimizing rule toward less volatility and more dispersion. Below s^* the rule prices volatility too low, so moving along the frontier saves more volatility than society loses in dispersion, and the opposite above s^* . The argument never solves for signal weights, so it covers every signal process in Assumption 2.

Figure 2 draws the frontier for the contest of Example 2, with volatility on the horizontal axis and dispersion on the vertical one. Along the frontier the slope is $-s$ (Lemma A.5), so the tangent at the rational point has slope $-(1 - \alpha)$ and the tangent at the team optimum has slope $-s^*$. A rule's loss relative to the team optimum is the vertical gap between the iso-loss line through it and the tangent at the optimum. The iso-loss line through the rational point cuts the frontier a second time, at $\bar{b}$.

Because $s = (1 - \alpha)b$ rises with b , welfare is quasi-concave in b with its peak at b^* , strictly so under Proposition 4(iii). Under that condition define the overshooting threshold as $\bar{b} := \sup\{b > b^* : L((1 - \alpha)b) < L(1 - \alpha)\} \in (b^*, \infty]$ when $b^* > 1$ and as $\bar{b} := \inf\{b < b^* : L((1 - \alpha)b) < L(1 - \alpha)\} \in [0, b^*)$ when $b^* < 1$; when $\bar{b}$ is finite and positive, continuity gives $L((1 - \alpha)\bar{b}) = L(1 - \alpha)$.

Corollary 2 (Welfare-improving and welfare-reducing biases). *Under the assumptions of Proposition 4(iii):*

- (i) If $\alpha > \alpha^*$, so that $b^* > 1$, a bias b raises welfare above its level under rationality if and only if $1 < b < \bar{b}$, where $\bar{b} > b^*$; welfare is strictly increasing in b on $(0, b^*]$ and strictly decreasing on $[b^*, \infty)$; and every $b < 1$ lowers welfare.

(ii) If $\alpha < \alpha^*$, so that $b^* < 1$, a bias b raises welfare if and only if $\bar{b} < b < 1$, where $\bar{b} < b^*$; welfare is strictly increasing in b on $(0, b^*]$ and strictly decreasing on $[b^*, \infty)$; and every $b > 1$ lowers welfare.

(iii) If $\alpha = \alpha^*$, every $b \neq 1$ lowers welfare.

The improving interval contains every factor between one and b^* and, by continuity, extends past b^* . Any debiasing from between one and b^* lowers welfare. Removing a bias beyond b^* raises welfare exactly when the bias exceeds $\bar{b}$, and the same bias under the same payoffs can lie on either side of $\bar{b}$, depending on signal precisions.

Example 2 (Two signals, continued). In the economy of Example 1, the family $k^{(s)}$ puts weight $\gamma(s) = \tilde{\tau}_z / (\tilde{\tau}_z + s\tau_v)$ on the public posterior mean and $1 - \gamma(s)$ on the private signal, and equations (23)–(24) of Angeletos and Pavan (2007) give $D(s) = (1 - \gamma(s))^2 / \tau_v$ and $V(s) = \gamma(s)^2 / \tilde{\tau}_z$. The loss (12) is a quadratic in the public weight,

$$L(s) = \frac{1}{2} \left[\frac{-W_{\sigma\sigma}}{\tau_v} + \frac{-W_{KK}}{\tilde{\tau}_z} \right] (\gamma(s) - \gamma^*)^2, \quad \gamma^* = \gamma(s^*),$$

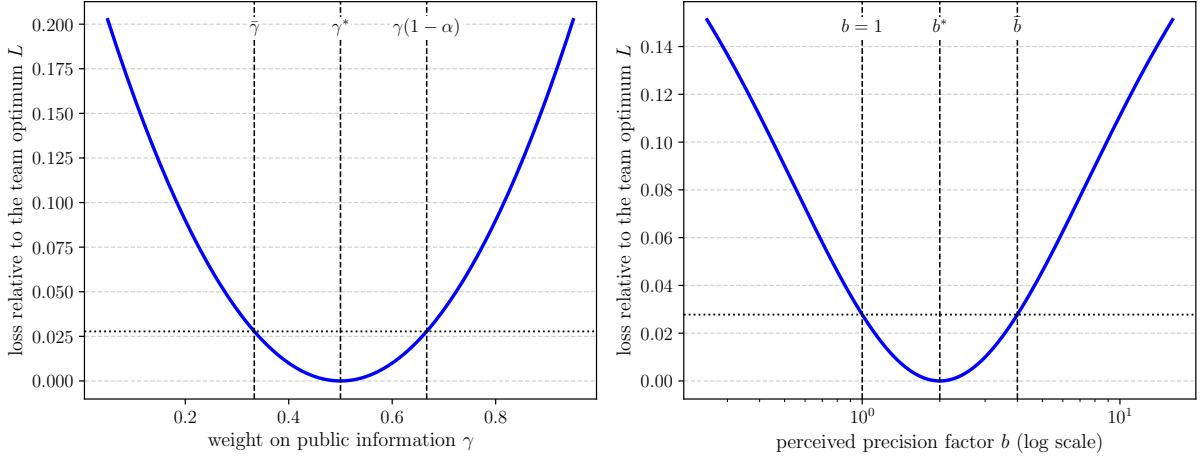
symmetric around the efficient weight γ^* . The overshooting threshold is the b at which the public weight is as far from γ^* on the other side as it is under rationality: $\gamma((1 - \alpha)\bar{b}) = 2\gamma^* - \gamma(1 - \alpha)$, provided the right-hand side lies in $(0, 1)$; otherwise $\bar{b}$ is ∞ or 0. Welfare is symmetric in the weight, not in b . Because γ is a convex decreasing function of s , the band of welfare-improving overshooting is wider in b than the band of welfare-improving undershooting when $b^* > 1$, and narrower when $b^* < 1$. Figure 1 draws both panels for the beauty contest of Morris and Shin (2002) with $r = \frac{1}{2}$ and unit precisions: $\alpha = \frac{1}{2}$, $\alpha^* = 0$, $b^* = 2$, and $\bar{b} = 4$. The loss under rationality is 0.028; at $b = \frac{1}{2}$, the same bias in the wrong direction, it is 0.090.

The possibility of beneficial overreaction is already established by Bizzarri and d'Arienzo (2024, Proposition 3): a small departure from rational expectations helps when traders initially respond too little to private information. Corollary 2 asks how far a precision error can go before that gain disappears. Under the restrictions here, welfare rises up to the exactly corrective factor and falls thereafter, so the comparison with rationality has a single improving interval.

3.4 Direction

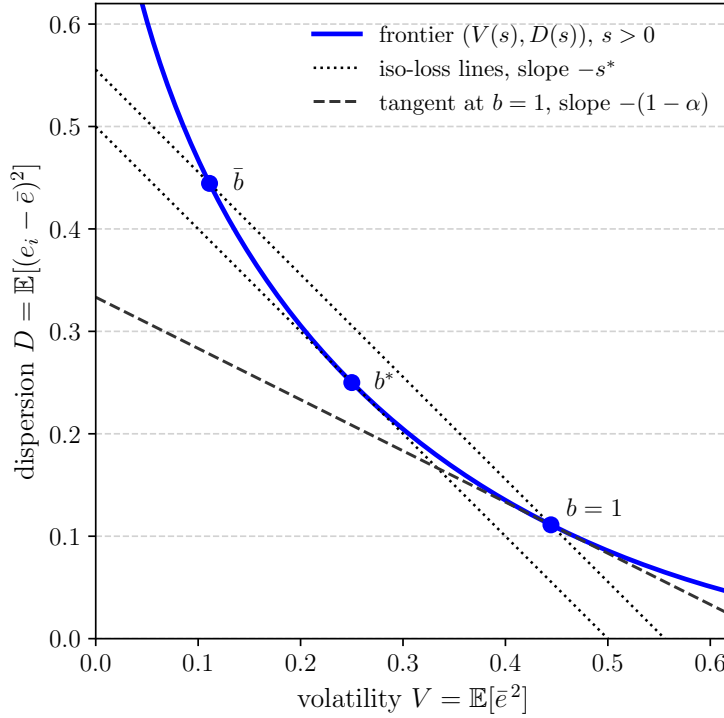
Overconfidence is corrective when private incentives favor coordination more than social welfare does; underconfidence is corrective when society values coordination more. Strategic complementarity alone does not determine the direction of correction. Agents may want to match others while coordinating either too much or too little relative to the efficient allocation. Overconfidence is corrective in the first case because it shifts decisions toward private information; underconfidence is corrective in the second because it increases reliance on common information. The following examples show how different social costs of dispersion and aggregate error produce these opposite prescriptions, even with the same private coordination motive. The three economies share the complete-information target θ , but reward matching, aggregate

Figure 1: Welfare loss relative to the team optimum in the two-signal example.



Notes: The left panel plots the loss against the weight on public information, the right panel against the precision factor b on a log scale. The payoff is $U = -(1-r)(k-\theta)^2 - r(k-K)^2 + r\sigma^2$ with $r = \frac{1}{2}$, so $\alpha = \frac{1}{2}$ and $\alpha^* = 0$; precisions are $\tau_v = \tilde{\tau}_z = 1$. Dashed lines mark rationality ($b = 1$), the corrective factor ($b^* = 2$), and the overshooting threshold ($\bar{b} = 4$), at public weights $\gamma(1-\alpha)$, γ^* and $\bar{\gamma}$ on the left; the dotted line is the loss under rationality.

Figure 2: The dispersion-volatility frontier in the two-signal example.



Notes: The solid curve is the pair $(V(s), D(s))$ traced by the family $k^{(s)}$ as s runs over $(0, \infty)$, in the economy of Figure 1: $\alpha = \frac{1}{2}$, $\alpha^* = 0$, $\tau_v = \tilde{\tau}_z = 1$. Dots mark the rational equilibrium ($b = 1$, $s = \frac{1}{2}$), the team optimum ($b^* = 2$, $s = 1$), and the overshooting threshold ($\bar{b} = 4$, $s = 2$). The dotted lines are iso-loss lines $D + s^*V = \text{const}$ at the social price $s^* = 1$, one tangent at the team optimum and one through the rational point; the loss (12) equals $\frac{1}{2}(-W_{\sigma\sigma})$ times the vertical distance between them. The dashed line is the tangent at the rational point, with slope $-(1-\alpha) = -\frac{1}{2}$.

accuracy, and low dispersion differently. I consider a beauty-contest game, a forecast economy, and an economy with a dispersion externality. The first two call for $b^* > 1$; the third calls for $b^* < 1$.

A private reward for following the crowd can draw too much weight away from individual information. In the contest of [Morris and Shin \(2002\)](#), $U = -(1-r)(k-\theta)^2 - r(k-K)^2 + r\sigma^2$, with $r \in (0,1)$. Each agent wants to stay close to the mean, but the average matching loss $-r\sigma^2$ cancels the last term in the payoff. Society consequently values only accuracy about θ . The coordination motives are $\alpha = r$ and $\alpha^* = 0$, so $b^* = 1/(1-r) > 1$ offsets the private incentive to follow common information. Dropping $r\sigma^2$ makes the matching loss a social cost as well: then $\alpha^* = \alpha$ and accurate beliefs are efficient.

Overconfidence can also improve aggregate accuracy when agents have no incentive to match one another. Consider forecasters with $U = -(k-\theta)^2 - \lambda(K-\theta)^2$, $\lambda > 0$. Each minimizes her own forecast error because she cannot affect the mean, so $\alpha = 0$. The planner also values the accuracy of that mean. Idiosyncratic errors average out, whereas errors induced by common information survive aggregation; the extra social value of aggregate accuracy therefore calls for more weight on private information. Here $\alpha^* = -\lambda$ and $b^* = 1 + \lambda$.

An additional social cost of dispersion reverses the correction, even when agents already want to coordinate. With $U = -(1-r)(k-\theta)^2 - r(k-K)^2 - d\sigma^2$, $r \in (0,1)$ and $d > 0$, each agent internalizes her own loss from departing from the mean but ignores the dispersion cost she imposes on others. The private motive remains $\alpha = r$; the social motive rises to $\alpha^* = (r+d)/(1+d)$. Underconfidence, $b^* = 1/(1+d) < 1$, induces more reliance on common information and reduces excessive dispersion. [Appendix B](#) verifies the assumptions for all three payoffs.

A small degree of overconfidence helps in the first two economies and hurts in the third. [Figure 3](#) displays this comparison at equal public and private precision. Define the information-related loss $\ell_b := D((1-\alpha)b) + (1-\alpha^*)V((1-\alpha)b)$, the dispersion and volatility costs in [\(11\)](#) after removing their common positive multiplier. Each panel plots ℓ_b/ℓ_1 , so values below one improve on rational expectations. The parameters illustrate the direction of the welfare effect.

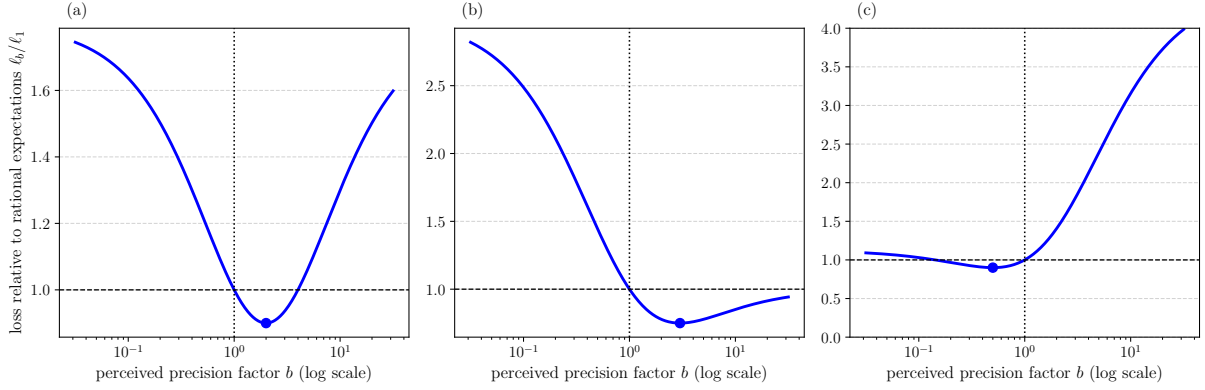
Payoffs determine the direction and the exactly corrective bias; signal precision determines how much overshooting remains beneficial. The two-signal example nevertheless gives a range of beneficial overconfidence that holds at every positive relative precision: when $b^* > 1$, every $1 < b \leq 2b^* - 1$ improves on rational expectations. To see the bound, let $q := \tau_v/\tilde{\tau}_z$. For $b^* \neq 1$, the fraction of the rational allocation's excess loss remaining at b is

$$\frac{\ell_b - \ell_{b^*}}{\ell_1 - \ell_{b^*}} = \left(\frac{b^* - b}{b^* - 1} \right)^2 \left(\frac{1 + (1-\alpha)q}{1 + (1-\alpha)bq} \right)^2. \quad (13)$$

The identity follows by expressing excess loss as a constant times $(\gamma_b - \gamma_{b^*})^2$ and canceling common factors. For $b > 1$, the second factor lies strictly between $1/b^2$ and one at every finite positive q . The first is at most one on $1 < b \leq 2b^* - 1$, which gives the stated improvement.

Better private information permits more overshooting in the overconfidence example, but

Figure 3: The direction of correction depends on the payoff externality.



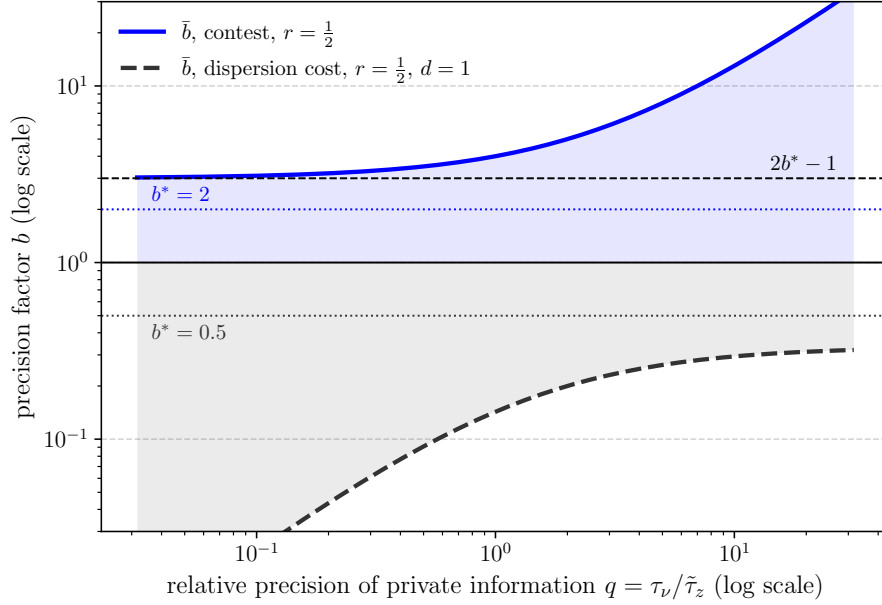
Notes: Each panel plots ℓ_b/ℓ_1 against the precision factor b on a log scale, with $\ell_b := D((1-\alpha)b) + (1-\alpha^*)V((1-\alpha)b)$ and $\tau_v = \tilde{\tau}_z = 1$. In panel (a), the payoff $-(1-r)(k-\theta)^2 - r(k-K)^2 + r\sigma^2$ rewards matching privately but offsets its average cost through $r\sigma^2$; $r = \frac{1}{2}$ gives $\alpha = \frac{1}{2}$, $\alpha^* = 0$ and $b^* = 2$. In panel (b), $U = -(k-\theta)^2 - \lambda(K-\theta)^2$ adds a social benefit from an accurate mean that each forecaster ignores; $\lambda = 2$ gives $\alpha = 0$, $\alpha^* = -\lambda$ and $b^* = 3$. In panel (c), $U = -(1-r)(k-\theta)^2 - r(k-K)^2 - d\sigma^2$ adds a dispersion cost that agents do not internalize; $r = \frac{1}{2}$ and $d = 1$ give $\alpha^* = \frac{3}{4}$ and $b^* = \frac{1}{2}$. The horizontal dashed line is the rational benchmark, the vertical dotted line marks $b = 1$, and dots mark optima.

less in the underconfidence example. The threshold in the two-signal class is

$$\bar{b} = \frac{2b^* - 1 + b^*(1-\alpha)q}{1 + (2-b^*)(1-\alpha)q}, \quad (14)$$

with $\bar{b} = \infty$ when the denominator is nonpositive and $\bar{b} = 0$ when the numerator is nonpositive. For $b^* > 1$ and a positive denominator, $\bar{b} - (2b^* - 1) = 2(1-\alpha)q(b^* - 1)^2$ divided by that denominator is positive. In the contest of Figure 4, $\bar{b} = 3 + q$, and the improving band approaches the bound $2b^* - 1 = 3$ as private precision vanishes. In the dispersion-cost economy, $\bar{b} = q/(4 + 3q)$ rises from zero toward one third; the improving band $(\bar{b}, 1)$ is widest when private information is poor. Signal precision changes these bands while leaving b^* fixed.

Figure 4: The band of welfare-improving precision factors as relative precision varies.



Notes: Each curve is the threshold $\bar{b}$ at which the loss returns to its level under rationality, plotted against $q = \tau_v / \tilde{\tau}_z$, both on log scales. Shaded areas are the factors that raise welfare above rationality: $1 < b < \bar{b}$ in the contest with $r = \frac{1}{2}$ ($b^* = 2$), and $\bar{b} < b < 1$ in the dispersion-cost economy with $r = \frac{1}{2}$ and $d = 1$ ($b^* = \frac{1}{2}$). Dotted lines mark b^* , the solid line marks $b = 1$, and the dashed line is the bound $2b^* - 1 = 3$ from (13) for the contest.

4 When actions produce information

Actions that respond more strongly to private information can help others learn from aggregate activity. This social return calls for a stronger early correction than the static problem requires. In a two-period economy with noisy observations of aggregate activity, the linear team optimum uses a larger precision factor at time 1 and returns to b^* at time 2. A constant error cannot reproduce that allocation, but sufficiently mild permanent overconfidence still raises welfare when private and social coordination motives coincide.

Learning from aggregate activity makes future information depend on today's decision rules. The planner must choose those rules jointly: changing the first rule changes the information available for the second. This learning externality also motivates the analysis of Vives (1997). The finite horizon allows me to solve the feedback from decisions to information by backward induction and isolate its effect on the corrective bias.⁵

⁵With observations of aggregate actions, the signal process depends on agents' policies. Changing those policies therefore changes future information as well as current allocations, so the planner cannot optimize separately at each time while holding information fixed. An infinite-horizon extension requires jointly characterizing decision rules and the evolution of information. Two periods provide a terminal problem to which the exogenous-information results apply conditional on the initial rule.

4.1 The efficient correction changes over time

The first-period response to private information determines the quality of the second-period signal. Let θ have a mean-zero Gaussian prior with precision $\tau_\theta > 0$. At time 1, agent i observes $x_i = \theta + v_i$, with independent idiosyncratic noise of precision $\tau_1 > 0$. At time 2 she retains x_i and observes $m_i = K_1 + u_i$, with independent idiosyncratic noise of precision $\tau_m > 0$. Each period she plays the same coordination game satisfying Assumptions 1 and 3, normalized to $\kappa = \kappa^* = \theta$. Payoffs add across periods without discounting, $\beta = 1$.

Each agent chooses her first-period action solely to maximize her expected first-period payoff. She is too small to affect K_1 , so her choice cannot change the information anyone receives at time 2. Her first-period action also has no direct effect on her second-period payoff. She therefore chooses her actions separately at each time, using x_i at time 1 and (x_i, m_i) at time 2, with beliefs computed under her perceived model. Both choices satisfy (3). The planner, by contrast, changes the decision rules of all agents together and therefore accounts for their effect on K_1 and subsequent learning.

The planner internalizes the learning benefit by choosing the first-period rule after solving the final-period team problem. I restrict the first rule to be symmetric and linear, $k_{i1} = wx_i$, and leave the second rule unrestricted; the resulting allocation is the linear team optimum. The first rule gives $K_1 = w\theta$. For $w \neq 0$, the observation m_i/w has precision $w^2\tau_m$ about θ , so a stronger early response raises later signal quality. Conditional on w , the second-period signals are exogenous and Gaussian, and Proposition 2 gives the efficient final rule. Substituting its welfare into the planner's objective reduces the joint problem to the choice of w . With $s^* := 1 - \alpha^*$ and $b^* := s^*/(1 - \alpha)$, the planner minimizes a positive multiple of

$$J(w) = \underbrace{\frac{w^2}{\tau_1} + s^* \frac{(1-w)^2}{\tau_\theta}}_{\text{time-1 loss}} + \underbrace{\frac{s^*}{\tau_\theta + s^*(\tau_1 + w^2\tau_m)}}_{\text{efficient time-2 loss}}. \quad (15)$$

The planner puts more weight on early private information than the static correction induces. At the minimum of the first term in $J(w)$, a small increase in w has only a second-order current cost, while the second term falls to first order as later information improves. The following proposition gives the resulting correction and shows why one factor cannot work in both periods.

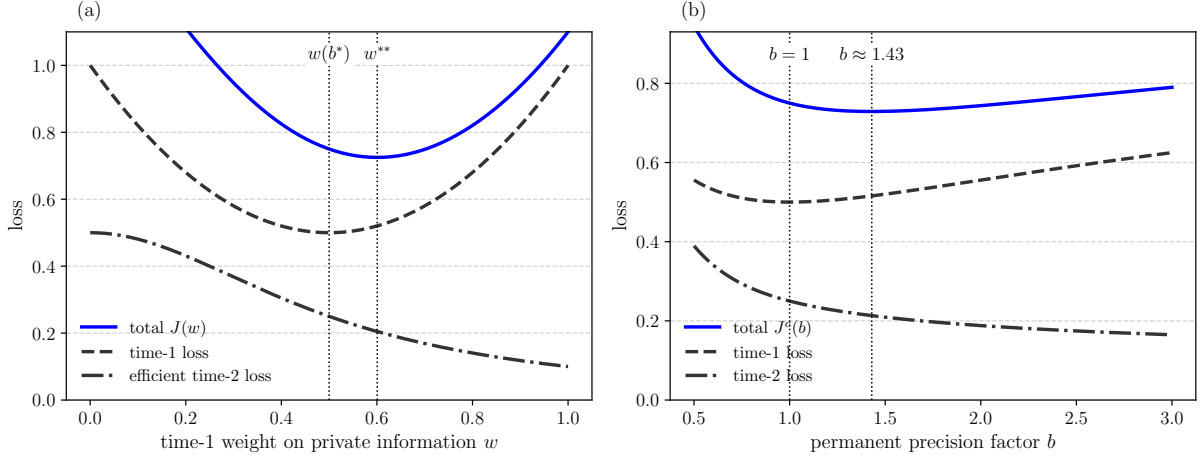
Proposition 5 (Front-loaded correction). *Let $\tau_m > 0$, let w^{**} be a minimizing time-1 weight in the linear team optimum, and let*

$$w(b) = \frac{(1 - \alpha)b\tau_1}{\tau_\theta + (1 - \alpha)b\tau_1}$$

be the time-1 equilibrium weight under bias b . Then:

- (i) $w(b^*) < w^{**} < 1$.
- (ii) *No constant bias implements the linear team optimum, including any pair of shock-specific precision distortions held fixed across periods.*

Figure 5: Early information use trades off current loss against later learning.



Notes: Parameters are $\alpha = \alpha^* = 0$, $\tau_\theta = \tau_1 = 1$, $\tau_m = 8$. Panel (a) plots the planner's loss (15) against the time-1 weight w , together with the time-1 loss $w^2/\tau_1 + s^*(1-w)^2/\tau_\theta$ and the efficient time-2 loss $s^*/(\tau_\theta + s^*(\tau_1 + w^2\tau_m))$; dotted lines mark the static optimum $w(b^*) = \frac{1}{2}$ and the linear team optimum $w^{**} = 0.60$. Panel (b) plots the constant-bias loss (17) against b with its two terms; dotted lines mark rationality and the best constant factor $b = 1.43$.

- (iii) A time-1 bias b_1 with $w(b_1) = w^{**}$, followed at time 2 by b^* on both idiosyncratic shocks, implements that optimum. The required b_1 is finite and exceeds b^* .

The extra early correction reflects the learning benefit, regardless of the sign of the static coordination gap. Every optimal weight remains below one, so a finite $b_1 > b^*$ induces it. At time 2 there are no later learners, and the factor returns to b^* . Figure 5(a) shows the tradeoff: the total loss keeps falling beyond the minimum of the first-period loss, until the additional current cost matches the gain from better final-period information.

Implementing this profile requires time-varying beliefs about precision. A profile (b_1, b_2) assigns a factor to each period, and agents play the distorted equilibrium of that period's game. At time 2 they take the actual aggregate loading $w(b_1)$ as given and apply their current factor to the idiosyncratic shocks, including the noise in the retained signal x_i . They therefore revise its perceived precision and accept a loading that their current model would not have generated at time 1. The profile describes naive, time-varying beliefs. It does not follow from Bayesian updating under a single fixed subjective model.

4.2 Welfare under a permanent bias

A permanent precision error can improve welfare even though it cannot attain the linear team optimum. The clearest case has no static coordination gap: accurate beliefs use current information efficiently, but a small permanent overconfidence makes early activity more informative. The learning gain is first order; the cost of distorting information use is second order. I evaluate this comparison under a single fixed perceived model.

A common factor b fixes perceived precision in both periods and determines the information that agents actually receive. It applies to the idiosyncratic noises v_i and u_i , with the prior

correct. Agents correctly anticipate the aggregate loading $w(b)$, and the true second-period private precision is $p(b) := \tau_1 + w(b)^2 \tau_m$. Proposition 3 then gives each period's equilibrium rule. Define its normalized one-period loss at effective scale s and true private precision p as

$$Q(s, p) := \frac{s^2 p + s^* \tau_\theta}{(\tau_\theta + s p)^2}. \quad (16)$$

The numerator collects dispersion and aggregate-error costs. Evaluating this loss at the two equilibrium information sets gives

$$J^c(b) = Q((1 - \alpha)b, \tau_1) + Q((1 - \alpha)b, \tau_1 + w(b)^2 \tau_m). \quad (17)$$

The second term uses the actual distorted final-period rule. In contrast, $J(w)$ in (15) uses the efficient final rule conditional on w .

Corollary 3 (A permanent bias above the static correction helps). *In the two-period economy with $\tau_m > 0$, J^c is strictly decreasing on $(0, b^*]$, and at b^**

$$\left. \frac{dJ^c(b)}{db} \right|_{b=b^*} = - \frac{2(s^*)^2 \tau_m w(b^*) w'(b^*)}{[\tau_\theta + s^* (\tau_1 + w(b^*)^2 \tau_m)]^2} < 0. \quad (18)$$

Hence every sufficiently small permanent increase above b^ improves two-period welfare relative to the constant bias b^* , and no constant factor at or below b^* is best. If $\alpha \geq \alpha^*$, every constant factor in $(1, b^*]$ improves on rational expectations. If $\alpha = \alpha^*$, then $b^* = 1$: sufficiently mild permanent overconfidence improves on rational expectations despite static efficiency.*

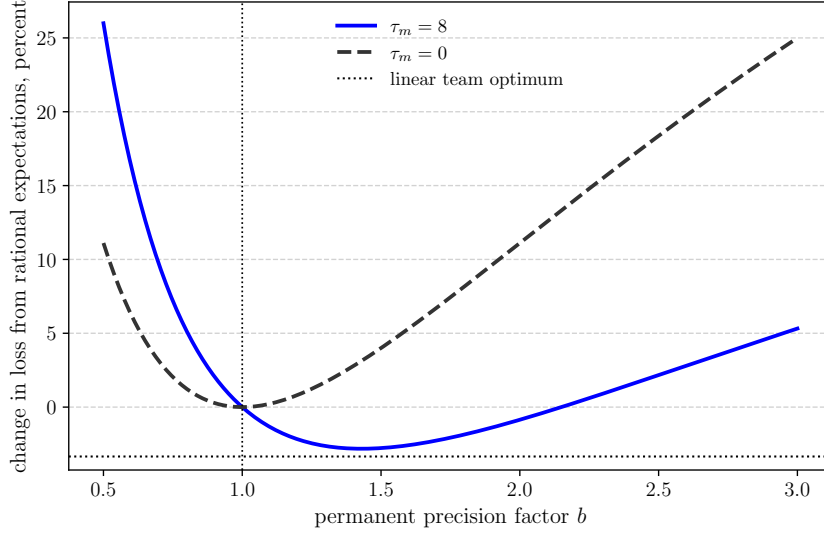
The learning benefit makes a permanent increase above b^* desirable at the margin. At b^* , each period's rule is efficient conditional on its available information, so a small increase in the factor has no first-order allocative cost at fixed signal precision. It does raise $w(b)$ and thus improves the second-period signal. The benefit from actions that reveal private information also underlies Bernardo and Welch (2001)'s result on overconfidence in sequential social learning. Here the derivative in (18) isolates that benefit at the static corrective factor.

The comparison with rational expectations depends on where the static correction lies. For $\alpha \geq \alpha^*$, loss falls from $b = 1$ to b^* and continues falling just beyond it. For $\alpha < \alpha^*$, learning can still make overconfidence beneficial, but the sign at $b = 1$ depends on parameters. For example, with $\alpha = 0$, $\alpha^* = 0.2$, $\tau_\theta = \tau_1 = 1$ and $\tau_m = 50$, the derivative of J^c at $b = 1$ is -0.006 . Mild overconfidence then helps even though the static correction calls for underconfidence, $b^* = 0.8$.

The numerical illustration separates the learning gain from any static coordination inefficiency. Figures 5(b) and 6 set $\alpha = \alpha^* = 0$, $\tau_\theta = \tau_1 = 1$ and $\tau_m = 8$. The permanent-bias loss initially falls because the final-period gain exceeds the first-period cost, and reaches its minimum near $b = 1.43$. A time-varying correction does better by retaining the informative early actions without distorting final-period information use. Removing the observation of aggregate activity removes the learning gain; accurate beliefs then minimize total loss.

The gain requires a stronger aggregate response to improve the signal-to-noise ratio. Here observation noise has fixed variance, so raising w makes m_i more informative. Exact observa-

Figure 6: Two-period loss under a permanent precision factor.



Notes: The figure plots the loss (17) as a percentage change from its value at $b = 1$, in an economy with no static coordination wedge: $\alpha = \alpha^* = 0$ and $\tau_\theta = \tau_1 = 1$. The solid curve has $\tau_m = 8$. The dashed curve has no time-2 observation of the aggregate. The dotted horizontal line is the linear team optimum with $\tau_m = 8$, implemented by the time-varying profile of Proposition 5(iii). Each curve is relative to its own rational benchmark.

tion of $K_1 = w\theta$ would instead reveal θ for any known nonzero w and leave no informational benefit from raising it. The conclusion therefore concerns noisy observation and the linear first-period rules allowed to the planner; it does not establish the optimal correction in a general dynamic economy.

5 Conclusion

I show when mistaken beliefs can fully correct the inefficient use of dispersed information. When decentralized decisions are efficient under complete information, a common error about relative information precision can reproduce the constrained-efficient allocation without changing incentives. Overprecision is corrective when private coordination incentives are excessive; underprecision is corrective when they are insufficient. I also characterize welfare over the full range of precision errors, identifying which incomplete corrections and overshooting errors remain beneficial. These results establish when restoring accurate beliefs would remove an existing correction to inefficient incentives and lower welfare.

Exact correction relies on restrictions on the economy and on the belief error. The analysis considers quadratic payoffs, Gaussian information, and a shared proportional misperception of idiosyncratic precision. It accommodates heterogeneous fundamentals and persistent signals, but does not establish an equivalent correction for arbitrary belief distortions or disagreement about signal quality. Extending the analysis to more general belief classes and repeated learning opportunities would allow a welfare assessment of a persistent bias as information accumulates. Allowing agents to revise their beliefs about signal quality would also connect the corrective effects studied here to the persistence of the errors themselves.

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A Proofs

A.1 Notation, conventions, and plan of the appendix

The main text uses the following equilibrium concept.

Definition (Equilibrium). A strategy is, for each period, a measurable function g_t from signal histories into actions with finite second moment, applied symmetrically by all agents: $k_{it} = g_t(x_i^t)$, $x_i^t := (x_{it}, x_{it-1}, \dots)$. A strategy induces the aggregates

$$K_t = \int_0^1 g_t(x_j^t) dj, \quad \sigma_t^2 = \int_0^1 \left(\tilde{k}_{jt} - \int_0^1 \tilde{k}_{lt} dl \right)^2 dj, \quad \tilde{k}_{jt} = g_t(x_j^t) - \kappa_1 \theta_{jt}.$$

An equilibrium is a strategy such that, for almost every signal history,

$$g_t(x_i^t) \in \arg \max_{k \in \mathbb{R}} \mathbb{E} [U(k, K_t, \sigma_t, \theta_{it}) \mid x_i^t],$$

where the conditional expectation is taken under the common prior of Assumption 2 and under the equilibrium strategy itself: agent i takes the processes (K_t, σ_t) generated by the others' adherence to g as given, because her own action does not affect them.

Throughout, t is fixed and suppressed. Let $\varepsilon_i = (\eta, v_i)$ denote the history of shocks up to t , with $\eta = (\eta_t, \eta_{t-1}, \dots)$ common and $v_i = (v_{it}, v_{it-1}, \dots)$ idiosyncratic. Under Assumption 2, η and v_i are independent Gaussian with covariances Σ_η and Σ_v (from here on the covariances of the stacked histories, block-diagonal across periods, each block the per-time covariance of Assumption 2), v_i is independent across agents, and I maintain the exact law of large numbers of Angeletos and Lian (2016) and Huo and Pedroni (2020), under the true measure and under every perceived measure introduced below: for any random variable $y_i = g(\eta, v_i)$ with finite second moment, $\int_0^1 y_j dj = \mathbb{E}[y_i \mid \eta]$ almost surely. In particular $\int v_j dj = 0$, $K = \mathbb{E}[k_i \mid \eta]$, and $\bar{\theta} = \mathbb{E}[\theta_i \mid \eta]$. Shocks have mean zero, so all constants are carried by κ_0 and κ_0^* . Payoffs are $U = U^0(k_i, K, \theta_i) + \frac{1}{2} U_{\sigma\sigma} \sigma^2$ with, by the exact law of large numbers, $\sigma^2 = \mathbb{E}[(\tilde{k}_i - \mathbb{E}[\tilde{k}_i \mid \eta])^2 \mid \eta]$ and $\tilde{k}_i = k_i - \kappa_1 \theta_i$, $\kappa_1 = -U_{k\theta} / U_{kk}$; $W_{\sigma\sigma} = U_{kk} + U_{\sigma\sigma}$ and $W_{KK} = U_{kk} + 2U_{kK} + U_{KK}$. Throughout, U_k^0, U_K^0 denote partial derivatives of the mean-action part.

Fundamentals. The fundamental splits into a common and an idiosyncratic part,

$$\theta_i = a'_\eta \eta + a'_v v_i, \quad \bar{\theta} = a'_\eta \eta,$$

where $a = (a_\eta, a_v)$ is the coefficient sequence $\psi_t(L)$ of Assumption 2, stacked over periods and split into the block that multiplies the common shocks and the block that multiplies the idiosyncratic ones; the expression for $\bar{\theta}$ is the exact law of large numbers (averaging kills v_i).

Signals. For $N \geq 0$ let $x_i^N = (x_{it}, \dots, x_{it-N})$ be the truncated signal history and x_i^∞ the full history. By Assumption 2 each signal is a linear combination of the shocks with square-summable coefficients. Stacking coefficients row by row,

$$x_i^N = B_{N\eta} \eta + B_{Nv} v_i,$$

where $B_{N\eta}$ and $B_{N\nu}$ have one row per signal kept in x_i^N , so finitely many rows, and one column per shock. When the economy is static, or more generally when only finitely many shocks and signals are relevant at a time, these are ordinary matrices. When the process has infinite memory (an AR(1) fundamental, say, whose moving-average representation loads on the entire past), the matrices have countably infinitely many columns. The blocks $B_{N\eta}$, $B_{N\nu}$ are the only way the matrices are ever used. $\text{Var}(x_i^N)$ is nonsingular for every N (redundant signals are dropped) and Σ_ν is positive definite.

Perceived measure. For $s > 0$ let P_s denote the Gaussian measure under which Σ_ν is replaced by Σ_ν/s , everything else unchanged, and $\mathbb{E}^{(s)}[\cdot \mid \cdot]$ conditional expectation under P_s ; $P = P_1$ is the true measure. For a linear combination $y = w'\varepsilon_i$ of the shocks, independence of η and ν_i gives

$$\text{Var}_{P_s}(y) = w'_\eta \Sigma_\eta w_\eta + \frac{1}{s} w'_\nu \Sigma_\nu w_\nu.$$

Both terms on the right are nonnegative, and the factors 1 and $1/s$ multiplying them each lie between $\min\{1, 1/s\}$ and $\max\{1, 1/s\}$; hence

$$\min\left\{1, \frac{1}{s}\right\} \text{Var}_P(y) \leq \text{Var}_{P_s}(y) \leq \max\left\{1, \frac{1}{s}\right\} \text{Var}_P(y). \quad (19)$$

This inequality is used for two purposes: (a) a linear combination of signals with positive variance under the true measure has positive variance under every perceived measure, so the signal variance matrix stays nonsingular when the measure changes (used in Lemma A.1); (b) in the extension to infinite histories, Section A.7, it transfers L^2 limits between measures.

Strategies. A strategy is a measurable function of the signal history with finite second moment; $\mathcal{H}$ denotes the space of such random variables under the true measure, and $\mathcal{H}_N \subset \mathcal{H}$ those measurable with respect to x_i^N . In the main line of every proof only finite histories appear; the structure of $\mathcal{H}$ matters only for the passage $N \rightarrow \infty$, and everything infinite-dimensional is collected in Section A.7. The target of the complete-information action is written, for a generic triple $(\kappa_0, \kappa_1, \kappa_2)$, as

$$\kappa(\theta_i, \bar{\theta}) - \kappa_0 = \kappa_1 \theta_i + \kappa_2 \bar{\theta} = c' \varepsilon_i, \quad c = (c_\eta, c_\nu) = ((\kappa_1 + \kappa_2) a_\eta, \kappa_1 a_\nu). \quad (20)$$

A.2 Two lemmas on linear forecasts

Lemma A.1. Fix $N, s > 0$, and a square-summable $w = (w_\eta, w_\nu)$. Then $\mathbb{E}^{(s)}[w' \varepsilon_i \mid x_i^N] = h' x_i^N$, where h is the unique solution of

$$h' G_s^N = w'_\eta \Sigma_\eta B'_{N\eta} + \frac{1}{s} w'_\nu \Sigma_\nu B'_{N\nu}, \quad G_s^N := \text{Var}_{P_s}(x_i^N) = B_{N\eta} \Sigma_\eta B'_{N\eta} + \frac{1}{s} B_{N\nu} \Sigma_\nu B'_{N\nu}. \quad (21)$$

Proof. Under P_s the pair $(w' \varepsilon_i, x_i^N)$ is jointly Gaussian with finite second moments, so the conditional expectation is the linear projection $\text{Cov}_{P_s}(w' \varepsilon_i, x_i^N) \text{Var}_{P_s}(x_i^N)^{-1} x_i^N$. The covariance is $w'_\eta \Sigma_\eta B'_{N\eta} + w'_\nu (\Sigma_\nu/s) B'_{N\nu}$, and G_s^N is nonsingular because $\text{Var}_P(x_i^N)$ is and (19) holds. ■

Lemma A.2. Fix $N, b > 0$, a slope $a < 1$, and a square-summable target $c = (c_\eta, c_\nu)$. Let h solve, for

all realizations of x_i^N ,

$$h'x_i^N = \mathbb{E}^{(b)}[c'\varepsilon_i \mid x_i^N] + a \mathbb{E}^{(b)}[h'B_{N\eta}\eta \mid x_i^N]. \quad (22)$$

Then h is unique and

$$h'x_i^N = \mathbb{E}^{((1-a)b)}[\tilde{c}'\varepsilon_i \mid x_i^N], \quad \tilde{c} = \left(\frac{c_\eta}{1-a}, c_\nu \right). \quad (23)$$

Proof. By Lemma A.1 applied under P_b to $c'\varepsilon_i$ and to the common random variable $h'B_{N\eta}\eta$ (whose ν -loading is zero), (22) holds for all x_i^N iff

$$h'G_b^N = c'_\eta \Sigma_\eta B'_{N\eta} + \frac{1}{b} c'_\nu \Sigma_\nu B'_{N\nu} + a h' B_{N\eta} \Sigma_\eta B'_{N\eta}.$$

Moving the last term to the left and using $G_b^N = B_{N\eta} \Sigma_\eta B'_{N\eta} + \frac{1}{b} B_{N\nu} \Sigma_\nu B'_{N\nu}$,

$$h' \left[(1-a) B_{N\eta} \Sigma_\eta B'_{N\eta} + \frac{1}{b} B_{N\nu} \Sigma_\nu B'_{N\nu} \right] = c'_\eta \Sigma_\eta B'_{N\eta} + \frac{1}{b} c'_\nu \Sigma_\nu B'_{N\nu}.$$

The bracket equals $(1-a) G_{(1-a)b}^N$. Dividing by $1-a > 0$,

$$h' G_{(1-a)b}^N = \frac{c'_\eta}{1-a} \Sigma_\eta B'_{N\eta} + \frac{1}{(1-a)b} c'_\nu \Sigma_\nu B'_{N\nu},$$

which is (21) with $s = (1-a)b$ and $w = \tilde{c}$. Uniqueness follows from nonsingularity of $G_{(1-a)b}^N$. ■

A.3 Equilibrium and distorted equilibrium (Propositions 1 and 3)

Proposition 1 is Proposition 3 at $b = 1$, so I prove the latter. Under the perceived measure P_b , a b -distorted equilibrium is a strategy $k \in \mathcal{H}^b$, the space of signal-measurable random variables with finite second moment under P_b , satisfying (3) with $\mathbb{E}^{(b)}[\cdot \mid x_i^\infty]$ and $K = \int k_j dj = \mathbb{E}^{(b)}[k_i \mid \eta]$; uniqueness is uniqueness in $\mathcal{H}^b$.

Proof of Proposition 3. Step 1: finite history. Fix N and suppose agents condition on x_i^N . Define T_N on $\mathcal{H}_N^b$, the x_i^N -measurable strategies with finite second moment under P_b , by

$$T_N k = \kappa_0(1-\alpha) + \kappa_1 \mathbb{E}^{(b)}[\theta_i \mid x_i^N] + \alpha \mathbb{E}^{(b)}[\mathbb{E}^{(b)}[k_i \mid \eta] \mid x_i^N],$$

the right side of (3) when all agents play k ; the dispersion term does not appear because it does not depend on the agent's own action. An equilibrium on the N -history is exactly a fixed point of T_N . For any two strategies, $T_N k - T_N k' = \alpha \mathbb{E}^{(b)}[\mathbb{E}^{(b)}[k_i - k'_i \mid \eta] \mid x_i^N]$, and a conditional expectation is nonexpansive in the L^2 -norm ($\mathbb{E}[(\mathbb{E}[y \mid \mathcal{F}])^2] \leq \mathbb{E}[y^2]$, by conditional Jensen's inequality); applying this twice,

$$\|T_N k - T_N k'\|_{L^2(P_b)} \leq |\alpha| \|k - k'\|_{L^2(P_b)}, \quad |\alpha| < 1 \text{ (Assumption 1)},$$

so T_N is a contraction and has a unique fixed point. To identify it, apply Lemma A.2 with slope

$a = \alpha$, scale b , and target $c_o := (\kappa_1 a_\eta, \kappa_1 a_v)$, the loadings of $\kappa_1 \theta_i$. Its solution h_N satisfies, with $s := (1 - \alpha)b$,

$$h'_N x_i^N = \mathbb{E}^{(s)}[\tilde{c}'_o \varepsilon_i \mid x_i^N], \quad \tilde{c}_o = \left(\frac{\kappa_1}{1 - \alpha} a_\eta, \kappa_1 a_v \right) = ((\kappa_1 + \kappa_2) a_\eta, \kappa_1 a_v) = c,$$

using $\kappa_1/(1 - \alpha) = \kappa_1 + \kappa_2$ from (2). Hence $h'_N x_i^N$ satisfies (22) with target c_o , and $\hat{k}_N := \kappa_0 + \mathbb{E}^{(s)}[c' \varepsilon_i \mid x_i^N] = \kappa_0 + h'_N x_i^N$ satisfies, after restoring the constants ($\kappa_0 = \kappa_0(1 - \alpha) + \alpha \kappa_0$),

$$\hat{k}_N = \kappa_0(1 - \alpha) + \kappa_1 \mathbb{E}^{(b)}[\theta_i \mid x_i^N] + \alpha \mathbb{E}^{(b)}[\hat{K}_N \mid x_i^N], \quad \hat{K}_N := \mathbb{E}^{(b)}[\hat{k}_N \mid \eta] = \kappa_0 + h'_N B_{N\eta} \eta, \quad (24)$$

where the common part of $h'_N x_i^N$ is $h'_N B_{N\eta} \eta$ under both the true and the perceived measure, an identity in the coefficients. So $\hat{k}_N = T_N \hat{k}_N$: the unique N -history equilibrium is $\hat{k}_N = \mathbb{E}^{(s)}[\kappa(\theta_i, \bar{\theta}) \mid x_i^N]$, which is (9) for a finite history. Its realization on the actual signals is the same formula read under P ; for a finite history the laws of x_i^N under P and P_b are mutually absolutely continuous.

Step 2: full history, existence and uniqueness. Define T on $\mathcal{H}^b$ by the same formula with x_i^∞ in place of x_i^N . The same two-contraction estimate gives $\|Tk - Tk'\| \leq |\alpha| \|k - k'\|$, so T has a unique fixed point k^b , the unique b -distorted equilibrium.

Step 3: passage to the limit. Let $\hat{k} := \kappa_0 + \mathbb{E}^{(s)}[c' \varepsilon_i \mid x_i^\infty]$. By Lemma A.8(ii), $\hat{k}_N \rightarrow \hat{k}$ in $L^2(P_{s'})$ for every $s' > 0$, in particular under P_b . Take $N \rightarrow \infty$ in (24), in $L^2(P_b)$: the left side converges to $\hat{k}$; $\mathbb{E}^{(b)}[\theta_i \mid x_i^N] \rightarrow \mathbb{E}^{(b)}[\theta_i \mid x_i^\infty]$ by Lemma A.8(ii); and for the last term, $\hat{K}_N = \mathbb{E}^{(b)}[\hat{k}_N \mid \eta] \rightarrow \mathbb{E}^{(b)}[\hat{k} \mid \eta] =: \hat{K}$ because conditioning is a contraction, so

$$\|\mathbb{E}^{(b)}[\hat{K}_N \mid x_i^N] - \mathbb{E}^{(b)}[\hat{K} \mid x_i^\infty]\| \leq \|\hat{K}_N - \hat{K}\| + \|\mathbb{E}^{(b)}[\hat{K} \mid x_i^N] - \mathbb{E}^{(b)}[\hat{K} \mid x_i^\infty]\| \rightarrow 0.$$

Hence $\hat{k} = T\hat{k}$, so $\hat{k} = k^b$ by Step 2. Finally, the b -distorted allocation is this strategy applied to the actual signals, i.e. its realization under P , which is well defined and given by the same formula by Lemma A.8(i); this is (9). ■

Proof of Corollary 1. Dividing the entire perceived covariance by c_η leaves every Gaussian conditional mean of a linear target unchanged, because the scalar cancels between the target-signal covariance and the inverse signal covariance. The normalized blocks are Σ_η and $(c_v/c_\eta)\Sigma_v$. Best responses depend on beliefs only through these conditional means, so Proposition 3 applies with $b = c_\eta/c_v$. The argument covers nonlinear strategies as well: the contraction in the proof of Proposition 3 gives a unique equilibrium under the perceived covariances $(c_\eta \Sigma_\eta, c_v \Sigma_v)$, and the linear rule of Proposition 3 at $b = c_\eta/c_v$ solves its fixed point. ■

A.4 The planner's problem (Proposition 2)

Let $\mathcal{W}(k) := \mathbb{E}[U(k_i, K, \sigma, \theta_i)] = \mathbb{E}[U^0(k_i, K, \theta_i)] + \frac{1}{2} U_{\sigma\sigma} \mathbb{E}[(\tilde{k}_i - \mathbb{E}[\tilde{k}_i \mid \eta])^2]$ for $k \in \mathcal{H}$, with $K = \mathbb{E}[k_i \mid \eta]$. Write $\Lambda(k) := \mathbb{E}[U_K^0(k_i, K, \theta_i) \mid \eta] = \int U_K^0(k_j, K, \theta_j) dj$ for the average mean-

action externality, an η -measurable random variable,

$$\Lambda(k) = U_K(0) + (U_{kK} + U_{KK})K + U_{K\theta}\bar{\theta}, \quad (25)$$

and $\zeta(k) := \tilde{k}_i - \mathbb{E}[\tilde{k}_i | \eta] = k_i - K - \kappa_1(\theta_i - \bar{\theta})$ for the non-fundamental idiosyncratic component of the action, which has $\mathbb{E}[\zeta(k) | \eta] = 0$.

Lemma A.3 (Planner's problem). *Under Assumptions 1 and 2: (i) for $k, \delta \in \mathcal{H}$,*

$$\begin{aligned} \mathcal{W}(k + \delta) &= \mathcal{W}(k) + \mathbb{E}[\delta_i (U_k^0(k_i, K, \theta_i) + \Lambda(k) + U_{\sigma\sigma}\zeta(k))] \\ &\quad + \frac{1}{2} [W_{\sigma\sigma} \mathbb{E}(\delta_i - \bar{\delta})^2 + W_{KK} \mathbb{E}\bar{\delta}^2], \quad \bar{\delta} := \mathbb{E}[\delta_i | \eta]; \end{aligned} \quad (26)$$

(ii) $\mathcal{W}$ is strictly concave and coercive on $\mathcal{H}$, and has a unique maximizer k^* ; (iii) k^* is characterized by $\mathbb{E}[U_k^0(k_i^*, K^*, \theta_i) + \Lambda(k^*) + U_{\sigma\sigma}\zeta(k^*) | x_i^\infty] = 0$; (iv) equivalently, k^* is the unique solution in $\mathcal{H}$ of

$$k_i^* = \mathbb{E}[\kappa^*(\theta_i, \bar{\theta}) | x_i^\infty] + \alpha^* \mathbb{E}[K^* - \kappa^*(\bar{\theta}, \bar{\theta}) | x_i^\infty], \quad \alpha^* = 1 - \frac{W_{KK}}{W_{\sigma\sigma}}, \quad (27)$$

with $\kappa_0^* = -(U_k(0) + U_K(0))/W_{KK}$, $\kappa_1^* = \kappa_1$, and $\kappa_1^* + \kappa_2^* = -(U_{k\theta} + U_{K\theta})/W_{KK}$.

Proof. (i) Expand the quadratic U^0 exactly to second order at (k_i, K, θ_i) in the direction $(\delta_i, \bar{\delta}, 0)$, using $\int \delta_j dj = \bar{\delta}$:

$$U^0(k_i + \delta_i, K + \bar{\delta}, \theta_i) = U^0(k_i, K, \theta_i) + U_k^0 \delta_i + U_K^0 \bar{\delta} + \frac{1}{2} [U_{kk} \delta_i^2 + 2U_{kK} \delta_i \bar{\delta} + U_{KK} \bar{\delta}^2].$$

Take expectations. For the first-order term in $\bar{\delta}$, condition on η : $\mathbb{E}[U_K^0(k_i, K, \theta_i) \bar{\delta}] = \mathbb{E}[\bar{\delta} \mathbb{E}[U_K^0 | \eta]] = \mathbb{E}[\bar{\delta} \Lambda(k)] = \mathbb{E}[\delta_i \Lambda(k)]$, the last step because $\Lambda(k)$ is η -measurable and $\mathbb{E}[\delta_i | \eta] = \bar{\delta}$. For the second-order terms, $\mathbb{E}[\delta_i \bar{\delta}] = \mathbb{E}[\bar{\delta}^2]$ and $\mathbb{E}[\delta_i^2] = \mathbb{E}(\delta_i - \bar{\delta})^2 + \mathbb{E}\bar{\delta}^2$ for the same reason. For the dispersion term, $\zeta(k + \delta) = \zeta(k) + (\delta_i - \bar{\delta})$, so $\frac{1}{2} U_{\sigma\sigma} \mathbb{E}[\zeta(k + \delta)^2] - \frac{1}{2} U_{\sigma\sigma} \mathbb{E}[\zeta(k)^2] = U_{\sigma\sigma} \mathbb{E}[\zeta(k)(\delta_i - \bar{\delta})] + \frac{1}{2} U_{\sigma\sigma} \mathbb{E}(\delta_i - \bar{\delta})^2$, and $\mathbb{E}[\zeta(k) \bar{\delta}] = \mathbb{E}[\bar{\delta} \mathbb{E}[\zeta(k) | \eta]] = 0$, so the first-order part is $U_{\sigma\sigma} \mathbb{E}[\zeta(k) \delta_i]$. Collecting terms gives (26), with $U_{kk} + U_{\sigma\sigma} = W_{\sigma\sigma}$ on $\mathbb{E}(\delta_i - \bar{\delta})^2$.

(ii) The quadratic form in (26) is bounded above by $\frac{1}{2} \max(W_{\sigma\sigma}, W_{KK}) \mathbb{E}\delta_i^2 = -c_0 \|\delta\|^2$ with $c_0 = \frac{1}{2} \min(-W_{\sigma\sigma}, -W_{KK}) > 0$, since $\mathbb{E}(\delta_i - \bar{\delta})^2 + \mathbb{E}\bar{\delta}^2 = \mathbb{E}\delta_i^2$. Hence $\mathcal{W}$ is a continuous, strongly concave quadratic functional on the Hilbert space $\mathcal{H}$; it is coercive and has a unique maximizer.

(iii) By (ii), k^* is the maximizer iff the linear term in (26) vanishes for all $\delta \in \mathcal{H}$, i.e. iff

$$\mathbb{E}[\delta_i (U_k^0(k_i^*, K^*, \theta_i) + \Lambda(k^*) + U_{\sigma\sigma}\zeta(k^*))] = 0$$

for all x_i^∞ -measurable δ with finite second moment, which is the stated conditional-expectation condition. With $U_{\sigma\sigma} = 0$ this is equation (43) of Angeletos and Lian (2016); with a common fundamental it is equation (29) of Angeletos and Pavan (2007).

(iv) The argument runs in four short steps: characterize the complete-information optimum pointwise; average the characterization to get the aggregate coefficients; subtract the average to get the idiosyncratic coefficient; then compare the dispersed-information first-order condition

with the conditional expectation of the pointwise one.

Step (a): the complete-information optimum. Let $\kappa^*(\theta_i, \bar{\theta})$ be the complete-information optimum, the maximizer of $\mathcal{W}$ over strategies measurable with respect to ε_i ; by (ii) with x_i^∞ replaced by ε_i it exists, is unique, and is characterized by the pointwise condition

$$U_k^0(\kappa_i^*, \bar{\kappa}^*, \theta_i) + \Lambda^{\text{ci}} + U_{\sigma\sigma}\zeta_i^{\text{ci}} = 0, \quad (28)$$

with $\bar{\kappa}^* := \int \kappa_j^* dj = \kappa^*(\bar{\theta}, \bar{\theta})$, $\Lambda^{\text{ci}} = U_K(0) + (U_{kK} + U_{KK})\bar{\kappa}^* + U_{K\theta}\bar{\theta}$, and $\zeta_i^{\text{ci}} = \kappa_i^* - \bar{\kappa}^* - \kappa_1(\theta_i - \bar{\theta})$. Since U_k^0 is affine, (28) reads

$$U_k(0) + U_{kk}\kappa_i^* + U_{kK}\bar{\kappa}^* + U_{k\theta}\theta_i + \Lambda^{\text{ci}} + U_{\sigma\sigma}\zeta_i^{\text{ci}} = 0. \quad (29)$$

Step (b): aggregate coefficients. Averaging (29) over i (the ζ term averages to zero, and θ_i averages to $\bar{\theta}$) gives

$$W_{KK}\bar{\kappa}^* = -[U_k(0) + U_K(0)] - (U_{k\theta} + U_{K\theta})\bar{\theta},$$

so $\bar{\kappa}^*$ is affine in $\bar{\theta}$ with the stated κ_0^* and $\kappa_1^* + \kappa_2^*$.

Step (c): idiosyncratic coefficient. Subtracting the average from (29) leaves only the deviations from means,

$$W_{\sigma\sigma}(\kappa_i^* - \bar{\kappa}^*) + U_{k\theta}(\theta_i - \bar{\theta}) - U_{\sigma\sigma}\kappa_1(\theta_i - \bar{\theta}) = 0,$$

i.e. $\kappa_1^* = (-U_{k\theta} + U_{\sigma\sigma}\kappa_1)/W_{\sigma\sigma} = -U_{k\theta}(U_{kk} + U_{\sigma\sigma})/(U_{kk}W_{\sigma\sigma}) = \kappa_1$; this is where the definition of σ net of $\kappa_1\theta_i$ is used, and it is (5).

Step (d): the fixed point. Take $\mathbb{E}[\cdot \mid x_i^\infty]$ of (29) and subtract it from the condition in (iii), using (25) and $\zeta(k^*) - \zeta_i^{\text{ci}} = (k_i^* - \kappa_i^*) - (K^* - \bar{\kappa}^*)$: the terms in $U_k(0)$, $U_K(0)$, θ_i , $\bar{\theta}$ cancel, leaving

$$(U_{kk} + U_{\sigma\sigma})(k_i^* - \mathbb{E}[\kappa_i^* \mid x_i^\infty]) + (2U_{kK} + U_{KK} - U_{\sigma\sigma})\mathbb{E}[K^* - \bar{\kappa}^* \mid x_i^\infty] = 0,$$

which is (27) with $\alpha^* = -(2U_{kK} + U_{KK} - U_{\sigma\sigma})/W_{\sigma\sigma} = 1 - W_{KK}/W_{\sigma\sigma}$, equation (10) of Angeletos and Pavan (2007). Conversely, (27) implies the condition in (iii) by reversing the steps, so its solution in $\mathcal{H}$ is unique. ■

Proof of Proposition 2. Let $s^* := 1 - \alpha^* = W_{KK}/W_{\sigma\sigma} = (-W_{KK})/(-W_{\sigma\sigma}) > 0$ and $c^* := ((\kappa_1^* + \kappa_2^*)a_\eta, \kappa_1^*a_\nu)$, the loadings of $\kappa^*(\theta_i, \bar{\theta}) - \kappa_0^*$.

Step 1: the N -history planner. Let k_N^* maximize $\mathcal{W}$ over $\mathcal{H}_N$. By Lemma A.3(i)–(ii) restricted to $\mathcal{H}_N$, it exists, is unique, and satisfies $\mathbb{E}[U_k^0(k_{N,i}^*, K_N^*, \theta_i) + \Lambda(k_N^*) + U_{\sigma\sigma}\zeta(k_N^*) \mid x_i^N] = 0$; the argument of Lemma A.3(iv) with $\mathbb{E}[\cdot \mid x_i^N]$ gives

$$k_{N,i}^* = \mathbb{E}[\kappa^*(\theta_i, \bar{\theta}) \mid x_i^N] + \alpha^*\mathbb{E}[K_N^* - \bar{\kappa}^* \mid x_i^N].$$

Since $\mathcal{W}$ is strictly concave on $\mathcal{H}_N$, this condition is sufficient as well as necessary, so it is enough to exhibit an affine solution; the maximizer is then that solution. Writing $k_{N,i}^* = \kappa_0^* + h'x_i^N$ and $K_N^* = \kappa_0^* + h'B_{N\eta}\eta$, and expanding $\kappa^*(\theta_i, \bar{\theta}) - \alpha^*\bar{\kappa}^* = \kappa_0^*(1 - \alpha^*) + \kappa_1^*\theta_i + (\kappa_2^* - \alpha^*(\kappa_1^* +$

$\kappa_2^*)\bar{\theta}$, this is (22) with $b = 1$, $a = \alpha^*$, and target $c = ((1 - \alpha^*)(\kappa_1^* + \kappa_2^*)a_\eta, \kappa_1^*a_v)$. Lemma A.2 (which requires only $1 - \alpha^* > 0$) gives $\tilde{c} = c^*$ and therefore $k_N^* = \mathbb{E}^{(s^*)}[\kappa^*(\theta_i, \bar{\theta}) \mid x_i^N]$. When the signal history is finite, $\mathcal{H}_N$ is the whole strategy space and the proof ends here.

Step 2: convergence. Since $\bigcup_N \mathcal{H}_N$ is dense in $\mathcal{H}$ (Lemma A.8(iii)) and $\mathcal{W}$ is continuous and strongly concave with modulus c_0 (Lemma A.3(ii)), $\mathcal{W}(k^*) \geq \mathcal{W}(k_N^*) \geq \mathcal{W}(\Pi_N k^*) \rightarrow \mathcal{W}(k^*)$, where Π_N is the L^2 projection onto $\mathcal{H}_N$, and strong concavity gives $c_0 \|k_N^* - k^*\|^2 \leq \mathcal{W}(k^*) - \mathcal{W}(k_N^*) \rightarrow 0$. On the other hand $k_N^* = \mathbb{E}^{(s^*)}[\kappa^* \mid x_i^N]$ converges in $L^2(P)$ to $\mathbb{E}^{(s^*)}[\kappa^* \mid x_i^\infty]$ by Lemma A.8(ii). The two limits coincide, which is (8). ■

Remark. The proof uses $W_{KK} < 0$ and $W_{\sigma\sigma} < 0$ (Assumption 1) but not a lower bound on α^* : sufficiency of the planner's first-order condition comes from concavity, not from a contraction. A lower bound $\alpha^* > -1$, which would be needed to invoke the equilibrium representation of [Huo and Pedroni \(2020\)](#) for the planner's fixed point, is therefore not required.

A.5 Proof of Theorem 1

Under Assumption 3, $c^* = c$ and $\kappa_0^* = \kappa_0$, so Propositions 3 and 2 give $k^b = \mathbb{E}^{((1-\alpha)b)}[\kappa(\theta_i, \bar{\theta}) \mid x_i^\infty]$ and $k^* = \mathbb{E}^{(1-\alpha^*)}[\kappa(\theta_i, \bar{\theta}) \mid x_i^\infty]$. For $s > 0$ write $k^{(s)} := \mathbb{E}^{(s)}[\kappa(\theta_i, \bar{\theta}) \mid x_i^\infty]$ and let $\mathcal{X}_v$ denote the closed linear span of the idiosyncratic components $\{B_{Nv}v_i\}_N$ of the signal history, P_v the $L^2(P)$ projection onto it, and $\iota(k) := k_i - \mathbb{E}[k_i \mid \eta]$ the idiosyncratic component of a strategy. For a linear strategy $k_i = h'x_i^\infty$, $\iota(k) = h'B_vv_i \in \mathcal{X}_v$.

(i) Immediate: at $b = b^*$, $(1 - \alpha)b = 1 - \alpha^*$, so $k^{b^*} = k^{(1-\alpha^*)} = k^*$ as random variables, hence almost surely under P ; with a finite history the coefficient vectors coincide and the equality holds for every realization; and $K^{b^*} = \mathbb{E}[k^{b^*} \mid \eta] = K^*$.

(ii) I show: $k^{(s)} = k^{(s')}$ for some $s \neq s'$ iff $\iota(k^*) = \kappa_1 P_v(a'_v v_i)$, and in that case $k^{(s)} = k^*$ for all $s > 0$. The right side is κ_1 times the projection of the idiosyncratic component of θ_i on the idiosyncratic components of the signals, so the criterion in the statement, that the efficient rule uses idiosyncratic information beyond forecasting the own idiosyncratic fundamental, is $\iota(k^*) \neq \kappa_1 P_v(a'_v v_i)$. The idea: the normal equations depend on s only through the factor $1/s$ multiplying their idiosyncratic part, so two values of s share a solution exactly when that idiosyncratic part imposes no constraint, and then every s shares it. Let h be the linear rule of $k^{(s)}$ on x_i^∞ (coefficients in the closed span). The normal equations under P_s are: for every element y of the closed span of x_i^∞ , $\text{Cov}_{P_s}(c'\varepsilon_i - h'x_i^\infty, y) = 0$. Decompose $y = y_\eta + y_v$ into its common and idiosyncratic components and similarly $c'\varepsilon_i - h'x_i^\infty = (c'_\eta\eta - h'B_\eta\eta) + (c'_v v_i - h'B_v v_i)$. Since $\eta \perp v_i$ under every P_s and Cov_{P_s} of idiosyncratic components is $1/s$ times Cov_P ,

$$\text{Cov}_P(c'_\eta\eta - h'B_\eta\eta, y_\eta) + \frac{1}{s} \text{Cov}_P(c'_v v_i - h'B_v v_i, y_v) = 0 \quad \forall y.$$

If the same h satisfies this for s and $s' \neq s$, subtracting gives $\text{Cov}_P(c'_v v_i - h'B_v v_i, y_v) = 0$ for all $y_v \in \mathcal{X}_v$, i.e. $c'_v v_i - h'B_v v_i \perp \mathcal{X}_v$; since $h'B_v v_i \in \mathcal{X}_v$, this says $h'B_v v_i = P_v(c'_v v_i) = \kappa_1 P_v(a'_v v_i)$, i.e. $\iota(k^{(s)}) = \kappa_1 P_v(a'_v v_i)$. Conversely, if $\iota(k^{(s)}) = \kappa_1 P_v(a'_v v_i)$ then the idiosyncratic covariance term vanishes identically and the normal equations reduce to $\text{Cov}_P(c'_\eta\eta - h'B_\eta\eta, y_\eta) = 0$, which does not involve s , so h solves them for every $s > 0$. Applying this at $s = s^*$ gives the claim, with

the criterion stated on k^* . In the common-value case $a_v = 0$ and the criterion reads $\iota(k^*) \neq 0$: the efficient rule loads on some idiosyncratic shock. The subtraction step yields orthogonality to $\{y_v : y \in \mathcal{L}\}$, which is dense in $\mathcal{X}_v$, and extends to $\mathcal{X}_v$ by continuity (for a finite history this closure step is vacuous; $\mathcal{L}$ is defined in Section A.7).

(iii) Take the common-value structure with one private signal $x_i = \bar{\theta} + v_i$ and one public signal $z = \bar{\theta} + \zeta$, all Gaussian with positive precisions. The efficient rule puts weight

$$\frac{(\kappa_1 + \kappa_2) s^* \tau_v}{\tau_\theta + \tau_z + s^* \tau_v}$$

on x_i , which is nonzero because $\kappa_1 + \kappa_2 = \kappa_1 / (1 - \alpha) = -U_{k\theta} / (U_{kk}(1 - \alpha)) \neq 0$ by Assumption 1; so $\iota(k^*) \neq 0$ and, by (ii), b^* is the only b that works for this structure; a fortiori it is the only b that works for every structure. For the second claim, take the structure in which $x_i = (\theta_i, \bar{\theta})$ is observed without noise (this satisfies Assumption 2; choose the loadings so that $\text{Var}(\bar{\theta}) = a'_\eta \Sigma_\eta a_\eta > 0$ and $a'_v \Sigma_v a_v > 0$, which makes $\text{Var}(x_i)$ nonsingular, and use the single signal $\bar{\theta}$ in the common-value case). Then $\mathbb{E}^{(s)}[\cdot | x_i]$ applied to any affine function of $(\theta_i, \bar{\theta})$ returns that function for every s , so $k^b = \kappa(\theta_i, \bar{\theta})$ and $k^* = \kappa^*(\theta_i, \bar{\theta})$ for every b . They coincide iff $\kappa = \kappa^*$ as affine functions on the support of $(\theta_i, \bar{\theta})$, which is all of $\mathbb{R}^2$ under the nondegeneracy just imposed, i.e. iff Assumption 3 holds (in the common-value case, iff $\kappa_0 = \kappa_0^*$ and $\kappa_1 + \kappa_2 = \kappa_1^* + \kappa_2^*$, which with $\kappa_1 = \kappa_1^*$ is the same condition). Both claims range over the loadings ψ_t of the fundamental as well as over $M_t(L)$. With ψ_t held fixed the second claim can fail: if $\bar{\theta}_t \equiv 0$ and $\kappa_0 = \kappa_0^*$, then $c = c^*$ whatever $\kappa_2 - \kappa_2^*$, and b^* implements the efficient allocation for every $M_t(L)$. ■

A.6 Welfare (Section 3.3)

Lemma A.4 (Welfare decomposition, equation (11)). *Under Assumption 1, for every $k \in \mathcal{H}$, with $e_i = k_i - \kappa^*(\theta_i, \bar{\theta})$ and $\bar{e} = \mathbb{E}[e_i | \eta] = K - \bar{\kappa}^*$,*

$$\mathcal{W}(k) = \mathcal{W}^{\text{ci}} - \frac{1}{2}(-W_{\sigma\sigma})\mathbb{E}(e_i - \bar{e})^2 - \frac{1}{2}(-W_{KK})\mathbb{E}\bar{e}^2, \quad \mathcal{W}^{\text{ci}} := \mathbb{E}[U^0(\kappa_i^*, \bar{\kappa}^*, \theta_i) + \frac{1}{2}U_{\sigma\sigma}(\zeta_i^{\text{ci}})^2].$$

Proof. Apply the exact expansion of Lemma A.3(i) at the point κ^* (which is not in $\mathcal{H}$, but the expansion is an algebraic identity between random variables and only uses $\int e_j dj = \bar{e}$ and $\mathbb{E}[e_i | \eta] = \bar{e}$), in the direction e : $\mathcal{W}(k) = \mathcal{W}^{\text{ci}} + \mathbb{E}[e_i(U_k^0(\kappa_i^*, \bar{\kappa}^*, \theta_i) + \Lambda^{\text{ci}} + U_{\sigma\sigma}\zeta_i^{\text{ci}})] + \frac{1}{2}[W_{\sigma\sigma}\mathbb{E}(e_i - \bar{e})^2 + W_{KK}\mathbb{E}\bar{e}^2]$. The bracket vanishes identically by the definition of κ^* (Lemma A.3(iv)). Since $\kappa_1^* = \kappa_1$, $\zeta_i^{\text{ci}} = 0$ and $\zeta(k) = e_i - \bar{e}$, so $\text{Disp}(k) = \mathbb{E}[\sigma^2]$ and $\mathcal{W}^{\text{ci}} = \mathbb{E}[U^0(\kappa_i^*, \bar{\kappa}^*, \theta_i)]$. ■

Lemma A.5 (Frontier lemma). *Under Assumptions 1 and 2, for every $s > 0$ the functional $J_s(k) := \text{Disp}(k) + s \text{Vol}(k) = \mathbb{E}(e_i - \bar{e})^2 + s\mathbb{E}\bar{e}^2$ has a unique minimizer on $\mathcal{H}$, namely $k^{(s)} = \mathbb{E}^{(s)}[\kappa^*(\theta_i, \bar{\theta}) | x_i^\infty]$. Under Assumption 3 this is the $k^{(s)}$ of Section 3.1.*

Proof. J_s is the loss functional of Lemma A.4 for a fictitious payoff with the same U_{kk} , $U_{k\theta}$, $U_k(0)$, $U_{\sigma\sigma}$ (hence the same κ_1 , σ , and $W_{\sigma\sigma}$) and with W_{KK} replaced by $sW_{\sigma\sigma} < 0$, i.e. $J_s = \frac{2}{-W_{\sigma\sigma}}[\mathcal{W}_s^{\text{ci}} - \mathcal{W}_s(k)]$ for that payoff; any such U' with $U'_{kk} = sW_{\sigma\sigma} - U_{kk} - 2U_{kK}$ and

$(U'_K(0), U'_{K\theta})$ chosen so that its complete-information optimum is κ^* (this pins down $U'_K(0)$ and $U'_{K\theta}$ through $\kappa_0^* = -(U_K(0) + U'_K(0))/W'_{KK}$ and $\kappa_1^* + \kappa_2^* = -(U_{K\theta} + U'_{K\theta})/W'_{KK}$, while $\kappa_1'^* = \kappa_1 = \kappa_1^*$ automatically) has these properties. Lemma A.3 and the proof of Proposition 2 apply verbatim to U' , with $\alpha'^* = 1 - s$ and $1 - \alpha'^* = s > 0$; they yield a unique minimizer equal to $\mathbb{E}^{(s)}[\kappa^* \mid x_i^\infty]$. No restriction on the sign or size of α'^* is used. ■

Lemma A.6 (Monotonicity along the family). *Under Assumptions 1 and 2, V is nonincreasing and D is nondecreasing on $(0, \infty)$.*

Proof. Let $s_1 < s_2$ and write $D_j = D(s_j)$, $V_j = V(s_j)$. By Lemma A.5, $k^{(s_1)}$ minimizes J_{s_1} and $k^{(s_2)}$ minimizes J_{s_2} , so

$$D_1 + s_1 V_1 \leq D_2 + s_1 V_2, \quad D_2 + s_2 V_2 \leq D_1 + s_2 V_1.$$

Adding the two inequalities gives $(s_2 - s_1)(V_2 - V_1) \leq 0$, so $V_2 \leq V_1$; the first inequality then gives $D_2 - D_1 \geq s_1(V_1 - V_2) \geq 0$. ■

Lemma A.7 (Continuity). *Under Assumptions 1 and 2, $s \mapsto k^{(s)}$ is continuous from $(0, \infty)$ to $\mathcal{H}$, and V and D are continuous; under Assumption 3 so is L .*

Proof. Fix $s' > 0$ and $k \in \mathcal{H}$, and put $\delta := k - k^{(s')}$, $\bar{\delta} := \mathbb{E}[\delta_i \mid \eta]$. The functional $J_{s'}$ is quadratic and its first variation vanishes at its minimizer $k^{(s')}$ in every direction of the linear space $\mathcal{H}$ (Lemma A.3(iii) for the fictitious payoff of Lemma A.5), so the expansion of Lemma A.3(i) around $k^{(s')}$ is exact and has no linear term:

$$J_{s'}(k) - J_{s'}(k^{(s')}) = \mathbb{E}(\delta_i - \bar{\delta})^2 + s' \mathbb{E}\bar{\delta}^2 \geq \min\{1, s'\} \mathbb{E}\delta_i^2 = \min\{1, s'\} \|k - k^{(s')}\|^2,$$

using $\mathbb{E}\delta_i^2 = \mathbb{E}(\delta_i - \bar{\delta})^2 + \mathbb{E}\bar{\delta}^2$. Apply this at $k = k^{(s)}$ and rewrite the left side with $J_{s'} = J_s + (s' - s)\text{Vol}$:

$$\min\{1, s'\} \|k^{(s)} - k^{(s')}\|^2 \leq [J_s(k^{(s)}) - J_s(k^{(s')})] + (s' - s)[V(s) - V(s')] \leq |s' - s| |V(s) - V(s')|,$$

the last step because the first bracket is nonpositive, $k^{(s)}$ being the minimizer of J_s . For $s' \in [s/2, 2s]$, Lemma A.6 gives $0 \leq V(s'), V(s) \leq V(s/2)$, and $V(s/2)$ is finite because $k^{(s/2)}$ and κ^* have finite second moments; hence

$$\|k^{(s)} - k^{(s')}\|^2 \leq \frac{V(s/2)}{\min\{1, s/2\}} |s' - s| \rightarrow 0 \quad (s' \rightarrow s).$$

Continuity of V , D and L follows because Vol , Disp and J_{s^*} are continuous on $\mathcal{H}$. ■

Proof of Proposition 4. Write $D(s) = \text{Disp}(k^{(s)})$, $V(s) = \text{Vol}(k^{(s)})$, $J_s(k) = \text{Disp}(k) + s\text{Vol}(k)$, and note from Lemma A.4 and $-W_{KK} = s^*(-W_{\sigma\sigma})$ that $\mathcal{W}(k) = \mathcal{W}^{\text{ci}} - \frac{1}{2}(-W_{\sigma\sigma})J_{s^*}(k)$, so $L(s) = \frac{1}{2}(-W_{\sigma\sigma})[J_{s^*}(k^{(s)}) - J_{s^*}(k^{(s^*)})]$, which is (12).

(i) is Lemma A.6. Write $D_j = D(s_j)$, $V_j = V(s_j)$ and keep its two inequalities, $D_1 + s_1 V_1 \leq D_2 + s_1 V_2$ and $D_2 + s_2 V_2 \leq D_1 + s_2 V_1$ for $s_1 < s_2$.

(ii) $L(s^*) = 0$ by definition. Let $s_1 < s_2 \leq s^*$. Then $\frac{2}{-W_{\sigma\sigma}}[L(s_1) - L(s_2)] = (D_1 - D_2) + s^*(V_1 - V_2) \geq -s_2(V_1 - V_2) + s^*(V_1 - V_2) = (s^* - s_2)(V_1 - V_2) \geq 0$, using $D_1 - D_2 \geq -s_2(V_1 - V_2)$ from the second inequality in (i) and $V_1 \geq V_2$. Let $s^* \leq s_1 < s_2$. Then $\frac{2}{-W_{\sigma\sigma}}[L(s_2) - L(s_1)] = (D_2 - D_1) + s^*(V_2 - V_1) \geq s_1(V_1 - V_2) - s^*(V_1 - V_2) = (s_1 - s^*)(V_1 - V_2) \geq 0$, using $D_2 - D_1 \geq s_1(V_1 - V_2)$ from the first inequality.

(iii) J_s is strongly convex on $\mathcal{H}$ (Lemma A.3(ii) for the fictitious payoff), so its minimizer is unique and, if $k^{(s_1)} \neq k^{(s_2)}$, both inequalities in (i) are strict, whence $V_2 < V_1$ and the inequalities in (ii) are strict whenever $s_2 < s^*$, respectively $s_1 > s^*$; at $s_2 = s^*$ or $s_1 = s^*$ strictness reads $L(s) > 0$ for $k^{(s)} \neq k^*$. By the proof of Theorem 1(ii), $k^{(s)} = k^{(s')}$ for some pair $s \neq s'$ iff $\iota(k^*) = \kappa_1 P_v(a'_v v_i)$, in which case $k^{(s)} = k^*$ for all s and $L \equiv 0$; otherwise $k^{(s)} \neq k^{(s')}$ for all $s \neq s'$ and all inequalities are strict. ■

Proof of Corollary 2. Welfare under b is $\mathcal{W}(k^{(1-\alpha)b})$ and under rationality $\mathcal{W}(k^{(1-\alpha)})$; the map $b \mapsto s = (1 - \alpha)b$ is increasing, and b^* corresponds to s^* . Under Proposition 4(iii), L is continuous (Lemma A.7), strictly decreasing on $(0, s^*]$ and strictly increasing on $[s^*, \infty)$, with $L(s^*) = 0$.

(i) $b^* > 1$ means $s^* > 1 - \alpha$. For $s \in (1 - \alpha, s^*]$, $L(s) < L(1 - \alpha)$; for $s < 1 - \alpha$, $L(s) > L(1 - \alpha)$; for $s > s^*$, L increases continuously from $0 < L(1 - \alpha)$, so $\{s > s^* : L(s) < L(1 - \alpha)\} = (s^*, \bar{s})$ with $\bar{s} = \sup\{s > s^* : L(s) < L(1 - \alpha)\} \in (s^*, \infty]$, and $L(\bar{s}) = L(1 - \alpha)$ when $\bar{s} < \infty$. Translating with $\bar{b} = \bar{s}/(1 - \alpha)$ gives the statement, including $\bar{b} > b^*$ and the monotonicity claims. (ii) is symmetric with $s^* < 1 - \alpha$: $\{s < s^* : L(s) < L(1 - \alpha)\} = (\bar{s}, s^*)$ with $\bar{s} = \inf\{s < s^* : L(s) < L(1 - \alpha)\} \in [0, s^*)$, and $\bar{b} = \bar{s}/(1 - \alpha)$. (iii) $s^* = 1 - \alpha$: $L(s) > 0 = L(s^*)$ for all $s \neq s^*$. ■

Verification of Example 2. In the two-signal economy with prior $\theta \sim N(0, \tau_\theta^{-1})$, public signal z of precision τ_z and private signal x_i of precision τ_v , let $\tilde{z} := \tau_z z / (\tau_\theta + \tau_z)$ be the public posterior mean and $\tilde{\tau}_z := \tau_\theta + \tau_z$ its precision. Under P_s the private precision is $s\tau_v$, so $\mathbb{E}^{(s)}[\theta | x_i, z] = \gamma(s)\tilde{z} + (1 - \gamma(s))x_i$ with $\gamma(s) = \tilde{\tau}_z / (\tilde{\tau}_z + s\tau_v)$, and, with the normalization $\kappa_0 = 0, \kappa_1 + \kappa_2 = 1$ of Example 1 (Assumption 3 then gives the same for κ^*), $k^{(s)} = \gamma(s)\tilde{z} + (1 - \gamma(s))x_i$. Then $e_i - \bar{e} = (1 - \gamma)(x_i - \theta)$ and $\bar{e} = \gamma(\tilde{z} - \theta)$, so $D = (1 - \gamma)^2 / \tau_v$ and $V = \gamma^2 / \tilde{\tau}_z$, which are equations (23)–(24) of Angeletos and Pavan (2007) with their κ_1 set to one, $\sigma_x^2 = 1/\tau_v$ and $\sigma_z^2 = 1/\tilde{\tau}_z$. Hence $J_{s^*}(k^{(s)}) = (1 - \gamma)^2 / \tau_v + s^* \gamma^2 / \tilde{\tau}_z =: f(\gamma)$, a strictly convex quadratic in γ with minimizer $\gamma^* = \tilde{\tau}_z / (\tilde{\tau}_z + s^* \tau_v) = \gamma(s^*)$ and $f(\gamma) - f(\gamma^*) = [1/\tau_v + s^*/\tilde{\tau}_z](\gamma - \gamma^*)^2$. Since $L(s) = \frac{1}{2}(-W_{\sigma\sigma})[f(\gamma(s)) - f(\gamma^*)]$ and $s^*(-W_{\sigma\sigma}) = -W_{KK}$, this is the displayed L . For the beauty contest of Figure 1, $U_{kk} = -2$, $U_{kK} = 2r$, $U_{KK} = -2r$, $U_{\sigma\sigma} = 2r$ give $W_{\sigma\sigma} = W_{KK} = -2(1 - r)$, $\alpha = r$, $\alpha^* = 0$, and with $r = \frac{1}{2}$ and unit precisions $L = (\gamma - \gamma^*)^2$, $\gamma(s) = 1/(1 + s)$, $\gamma^* = \frac{1}{2}$, $\gamma(1 - \alpha) = \frac{2}{3}$, $\bar{\gamma} = \frac{1}{3}$, $b^* = 2$, $\bar{b} = 4$. Since γ is a strictly decreasing bijection from $(0, \infty)$ onto $(0, 1)$, L decreases up to s^* and increases thereafter, and $L(\bar{s}) = L(1 - \alpha)$ on the far side of s^* iff $\gamma(\bar{s}) = 2\gamma^* - \gamma(1 - \alpha)$, which has a solution in $(0, 1)$ iff $2\gamma^* - \gamma(1 - \alpha) \in (0, 1)$. Finally $\gamma''(s) = 2\tilde{\tau}_z \tau_v^2 / (\tilde{\tau}_z + s\tau_v)^3 > 0$, so γ is convex: equal intervals in γ correspond to longer intervals in s the further below γ^* they lie, which gives the stated asymmetry of the improvement bands. ■

A.7 Infinite signal histories

This section collects everything infinite-dimensional. It is used only in the limit steps flagged above (Step 3 of Proposition 3, Step 2 of Proposition 2, and one closure step in Theorem 1(ii)); when agents observe finitely many signals in total, the finite-history steps prove the propositions in full. Two issues arise with an infinite history. Convergence of conditional expectations given x_i^N to those given x_i^∞ is the martingale convergence theorem. The second issue is that the distorted equilibrium is computed under P_b but evaluated under P , and for an infinite history the laws of x_i^∞ under the two measures are mutually singular: two Gaussian measures are either equivalent or mutually singular (the Feldman-Hájek dichotomy), and with idiosyncratic noise arriving every period the time average of the squared standardized idiosyncratic innovations converges to a constant that differs under P and P_b by the factor $1/b$. A strategy defined only almost surely under P_b is then not defined under P . The remedy is to treat a linear strategy as a list of coefficients on signals. Let $\mathcal{L}_0$ be the set of finite linear combinations of signals and $\mathcal{L}$ its completion under the $L^2(P)$ -norm. By (19) the P - and P_s -norms are equivalent on $\mathcal{L}_0$, so $\mathcal{L}$ is the same space whichever norm is used to complete it, and each element of $\mathcal{L}$ has a realization as an $L^2(P_s)$ random variable for every $s > 0$, the limit of one and the same coefficient sequence. Applying a strategy to the actual signals means taking its P -realization, and convergence of linear strategies means convergence in $\mathcal{L}$, which is L^2 convergence under every P_s at once.

The same viewpoint explains the assertion, made in the definition of the b -distorted equilibrium, that the cross-sectional mean generated by a linear strategy is the same under the perceived and the true model. For a finite combination $y = h'x_i^N = h'B_{N\eta}\eta + h'B_{N\nu}\nu_i$, the cross-sectional mean is $h'B_{N\eta}\eta$, because averaging kills the idiosyncratic term whatever covariance is assigned to it: this is an identity in the coefficients, not a probabilistic statement, so it holds under P and under every P_s , and the mean is literally the same random variable. Lemma A.8(i) extends this to limits.

Lemma A.8 (Infinite-history toolkit). *Under Assumption 2:*

- (i) *Every element of $\mathcal{L}$ has a realization in $L^2(P_s)$ for each $s > 0$, obtained as the limit of the same sequence of finite combinations; the common part of an element of $\mathcal{L}$ is the same random variable under every P_s , and by the exact law of large numbers it is the cross-sectional mean the strategy generates.*
- (ii) *For any square-summable w and any $s > 0$, $\mathbb{E}^{(s)}[w'\varepsilon_i \mid x_i^N] \rightarrow \mathbb{E}^{(s)}[w'\varepsilon_i \mid x_i^\infty]$ as $N \rightarrow \infty$, in $L^2(P_{s'})$ for every $s' > 0$.*
- (iii) *$\bigcup_N \mathcal{H}_N$ is dense in $\mathcal{H}$.*

Proof. (i) The first claim is the norm equivalence (19) applied to Cauchy sequences in $\mathcal{L}_0$, as explained above. For the second, the map $y \mapsto \mathbb{E}_{P_s}[y \mid \eta]$ sends $h'x_i^N \in \mathcal{L}_0$ to $h'B_{N\eta}\eta$, a finite linear combination of common shocks that does not depend on s ; and since every P_s modifies only Σ_ν , all perceived measures agree on the law of the common shocks, so the codomain

$L^2(\sigma(\eta))$ is one fixed space independent of the measure. The map is an L^2 -contraction under each P_s , hence extends continuously to $\mathcal{L}$, and the extensions agree because they are limits of one and the same sequence. The identification of the common part with the cross-sectional mean is the exact law of large numbers, which is maintained under every perceived measure (Section A.1).

(ii) Under P_s , the σ -fields generated by x_i^N increase to the one generated by x_i^∞ , and $w'\varepsilon_i$ has finite second moment by square-summability, so $\mathbb{E}^{(s)}[w'\varepsilon_i | x_i^N] \rightarrow \mathbb{E}^{(s)}[w'\varepsilon_i | x_i^\infty]$ in $L^2(P_s)$ by the martingale convergence theorem. Each term of the sequence lies in $\mathcal{L}_0$ (Gaussian conditional expectations are linear), so the convergence is convergence in $\mathcal{L}$, which by (i) is $L^2(P_{s'})$ convergence of the corresponding realizations for every s' .

(iii) For $y \in \mathcal{H}$, $\mathbb{E}[y | x_i^N] \in \mathcal{H}_N$ and $\mathbb{E}[y | x_i^N] \rightarrow \mathbb{E}[y | x_i^\infty] = y$ in $L^2(P)$, again by martingale convergence. ■

B The economies of Section 3.4

The contest, forecast-aggregation, competitive, and dispersion-cost economies satisfy all three assumptions under the restrictions stated below. The investment-spillover economy has a complete-information inefficiency that no precision factor removes across information environments. All five have a common fundamental, so σ measures raw action dispersion.

B.1 The beauty contest of Morris and Shin

The primitive payoff is $u_i = -(1-r)(k_i - \theta)^2 - r(L_i - \bar{L})$ with $r \in (0, 1)$, where $L_i = \int (k_j - k_i)^2 dj$ is the mean square distance of others' actions from i 's and $\bar{L} = \int L_j dj$ its cross-sectional average. Using $L_i = (k_i - K)^2 + \sigma_k^2$ and $\bar{L} = 2\sigma_k^2$, where σ_k is the raw dispersion of actions,

$$U(k, K, \sigma, \theta) = -(1-r)(k - \theta)^2 - r(k - K)^2 + r\sigma^2 :$$

the σ^2 term is the part of the punished distance that everyone suffers equally, and it enters with a positive sign, an externality that makes alignment privately but not socially valuable. The derivatives are

$$U_{kk} = -2, \quad U_{kK} = 2r, \quad U_{KK} = -2r, \quad U_{\sigma\sigma} = 2r, \quad U_{k\theta} = 2(1-r),$$

so $\alpha = r \in (0, 1)$, $W_{KK} = -2 + 4r - 2r = -2(1-r) < 0$, $W_{\sigma\sigma} = -2 + 2r = -2(1-r) < 0$, and

$$\alpha^* = 1 - \frac{W_{KK}}{W_{\sigma\sigma}} = 0.$$

The complete-information action solves $k = (1-r)\theta + rK$, so $\kappa(\theta, \theta) = \theta$; and $\kappa^* = \theta$ as well (Step (b) of Lemma A.3(iv): $W_{KK}\bar{\kappa}^* = -(U_{k\theta} + U_{K\theta})\theta$ with $U_{K\theta} = 0$ gives $\bar{\kappa}^* = \theta$), so Assumption 3 holds. All of Assumption 1 is satisfied. Dropping the $r\sigma^2$ term changes the economy: then $U_{\sigma\sigma} = 0$, $W_{\sigma\sigma} = -2$, $W_{KK} = -2(1-r)$, and $\alpha^* = 1 - (1-r) = r = \alpha$, so the

contest without the externality is efficient and $b^* = 1$, the point made in Section 3.4.

B.2 The forecast-aggregation economy

Each agent wants only to be right, and society also cares about the accuracy of the average:

$$U(k, K, \sigma, \theta) = -(k - \theta)^2 - \lambda(K - \theta)^2, \quad \lambda > 0.$$

The derivatives are

$$U_{kk} = -2, \quad U_{kK} = 0, \quad U_{KK} = -2\lambda, \quad U_{\sigma\sigma} = 0, \quad U_{k\theta} = 2, \quad U_{K\theta} = 2\lambda,$$

so $\alpha = 0$, $W_{KK} = -2 - 2\lambda < 0$, $W_{\sigma\sigma} = -2 < 0$, and

$$\alpha^* = 1 - \frac{W_{KK}}{W_{\sigma\sigma}} = 1 - (1 + \lambda) = -\lambda.$$

The complete-information action solves $-2(k - \theta) = 0$, so $\kappa(\theta, \theta) = \theta$; and $\kappa^* = \theta$ as well, since Step (b) of Lemma A.3(iv) gives $W_{KK}\bar{\kappa}^* = -(U_{k\theta} + U_{K\theta})\theta = -2(1 + \lambda)\theta$, so Assumption 3 holds. Assumption 1 holds for every $\lambda > 0$, with $|U_{kK}/U_{kk}| = 0$. The planner's side of the argument does not need a lower bound on α^* (the remark after the proof of Proposition 2), so λ may exceed one.

B.3 The efficient competitive economy

In the incomplete-market production economy of Angeletos and Pavan (2007, Section 6.1), household i produces k_i of a good before demand θ is known, at cost $k_i^2/2$, and sells at the market-clearing price $p = \theta - \zeta K$, $\zeta > 0$; consuming the proceeds yields, after substituting demand, budget, and market clearing,

$$U(k, K, \sigma, \theta) = (\theta - \zeta K)k - \frac{1}{2}k^2 + \frac{1}{2}\zeta K^2 + e,$$

with e the endowment. The derivatives are $U_{kk} = -1$, $U_{kK} = -\zeta$, $U_{KK} = \zeta$, $U_{\sigma\sigma} = 0$, $U_{k\theta} = 1$, so $\alpha = -\zeta < 0$: actions are strategic substitutes, and Assumption 1 requires $\zeta < 1$ (the source paper needs no such bound because it does not use the equilibrium representation; nothing below depends on this difference). Then $W_{KK} = -1 - 2\zeta + \zeta = -(1 + \zeta) < 0$, $W_{\sigma\sigma} = -1 < 0$, and

$$\alpha^* = 1 - (1 + \zeta) = -\zeta = \alpha, \quad \kappa(\theta) = \kappa^*(\theta) = \frac{\theta}{1 + \zeta},$$

the last equalities as computed in the source paper; $\kappa = \kappa^*$ is the first welfare theorem at work. Assumptions 1 (for $\zeta < 1$) and 3 hold. Angeletos and Lian (2016, Section 9.3) report $\alpha = \alpha^*$ for a real-business-cycle economy; the same conclusion holds there if, in addition, $\kappa = \kappa^*$ and the payoff has the form of Assumption 1, which I do not verify here.

B.4 A dispersion externality

Let

$$U(k, K, \sigma, \theta) = -(1-r)(k-\theta)^2 - r(k-K)^2 - d\sigma^2, \quad 0 < r < 1, \quad d > 0.$$

An individual chooses her action taking dispersion as given, so

$$U_{kk} = -2, \quad U_{kK} = 2r, \quad U_{KK} = -2r, \quad U_{\sigma\sigma} = -2d,$$

and $\alpha = r$. The welfare curvatures are

$$W_{KK} = -2(1-r) < 0, \quad W_{\sigma\sigma} = -2(1+d) < 0.$$

Consequently,

$$\alpha^* = 1 - \frac{1-r}{1+d} = \frac{r+d}{1+d} > r.$$

The complete-information equilibrium satisfies $k = (1-r)\theta + rK$, hence $\kappa = \theta$. The planner's aggregate first-order condition gives

$$W_{KK}\kappa^* = -(U_{k\theta} + U_{K\theta})\theta = -2(1-r)\theta,$$

so $\kappa^* = \theta$ as well. The externality therefore leaves complete-information actions efficient and changes the desired response to dispersed information.

B.5 Investment spillovers: the excluded case

In the production-externality economy of [Angeletos and Pavan \(2007, Section 6.2\)](#), payoffs are

$$U(k, K, \sigma, \theta) = [(1-a)\theta + aK]k - \frac{1}{2}k^2, \quad a \in (0, \frac{1}{2}),$$

so $U_{kk} = -1$, $U_{kK} = a$, $U_{KK} = U_{\sigma\sigma} = 0$, $U_{k\theta} = 1-a$, giving $\alpha = a$, $W_{KK} = -1 + 2a$, $W_{\sigma\sigma} = -1$, $\alpha^* = 2a = 2\alpha$, and

$$\kappa(\theta) = \theta, \quad \kappa^*(\theta) = \frac{1-a}{1-2a}\theta \neq \kappa(\theta).$$

Assumption 1 holds, but Assumption 3 fails: the spillover distorts the complete-information equilibrium itself, since the social return to investment exceeds the private one state by state.

Summary

	α	α^*	$\kappa = \kappa^*$?	Ass. 1 and 3	b^*
Beauty contest (B.1)	r	0	yes	hold	$\frac{1}{1-r} > 1$
Forecast aggregation (B.2)	0	$-\lambda$	yes	hold	$1 + \lambda > 1$
Competitive economy (B.3)	$-\zeta$	$-\zeta$	yes	hold ($\zeta < 1$)	1
Dispersion cost (B.4)	r	$\frac{r+d}{1+d}$	yes	hold	$\frac{1}{1+d} < 1$
Investment spillovers (B.5)	a	$2a$	no	Ass. 3 fails	none

C Proofs for Section 4

C.1 Proof of Proposition 5

Setup and notation. Payoffs at each time are drawn from the class of Assumption 1 with a common fundamental and $\kappa = \kappa^* = \theta$ (normalization $\kappa_0 = 0$, $\kappa_1 + \kappa_2 = 1$), so Lemma A.4 applies period by period: welfare at a time equals its complete-information level minus $\frac{1}{2}(-W_{\sigma\sigma})[\text{Disp} + s^*\text{Vol}]$ with $s^* = 1 - \alpha^*$, and total welfare is the sum over the two times. Write $b^* = s^*/(1 - \alpha)$.

At time 1 the information is the private signal $x_i = \theta + v_i$ alone, and a symmetric linear rule is $k_{i1} = wx_i$. Then $\bar{e}_1 = (w - 1)\theta$ and $e_{i1} - \bar{e}_1 = wv_i$, so

$$J_1(w) := \text{Disp}_1 + s^*\text{Vol}_1 = \frac{w^2}{\tau_1} + s^*\frac{(1 - w)^2}{\tau_\theta},$$

a strictly convex function minimized at

$$w(b^*) = \frac{s^*\tau_1}{\tau_\theta + s^*\tau_1},$$

which is also, replacing s^* by $(1 - \alpha)b$, the time-1 weight $w(b)$ of the b -distorted equilibrium $\mathbb{E}^{((1-\alpha)b)}[\theta \mid x_i]$ (Proposition 3 with the prior as the only public information); $b \mapsto w(b)$ is a continuous, strictly increasing bijection from $(0, \infty)$ onto $(0, 1)$.

At time 2 the information is (x_i, m_i) with $m_i = K_1 + u_i = w\theta + u_i$. For $w \neq 0$, $m_i/w = \theta + u_i/w$ is a private signal of precision $w^2\tau_m$, independent of v_i , so the total private precision is $\tau_p(w) = \tau_1 + w^2\tau_m$. The time-2 team optimum given w is $\mathbb{E}^{(s^*)}[\theta \mid x_i, m_i]$ (Proposition 2). Under P_s , for any $s > 0$, both private precisions are multiplied by s , so the forecast weights the two private signals in their true precision ratio $\tau_1 : w^2\tau_m$ and depends on them only through the statistic $\hat{x}_i := (\tau_1 x_i + w\tau_m m_i)/\tau_p(w) = \theta + \hat{v}_i$, whose noise $\hat{v}_i = (\tau_1 v_i + w\tau_m u_i)/\tau_p(w)$ has precision $\tau_p(w)$:

$$\mathbb{E}^{(s)}[\theta \mid x_i, m_i] = \frac{s\tau_1 x_i + s w^2\tau_m (m_i/w)}{\tau_\theta + s\tau_1 + s w^2\tau_m} = \frac{s\tau_p(w)}{\tau_\theta + s\tau_p(w)} \hat{x}_i.$$

This is why a single true private precision $\tau_p(w)$ suffices, at $s = s^*$ here and at $s = (1 - \alpha)b$ in C.2. Evaluating $\text{Disp} + s^*\text{Vol}$ at the time-2 team optimum gives, by the same algebra as in the verification of Example 2 with the prior in the role of the public signal,

$$J_2^*(w) = \min_{\gamma} \left[\frac{(1 - \gamma)^2}{\tau_p(w)} + s^* \frac{\gamma^2}{\tau_\theta} \right] = \frac{s^*}{\tau_\theta + s^*\tau_p(w)},$$

which is strictly decreasing in w on $(0, \infty)$: $\frac{d}{dw} J_2^* = -(s^*)^2 \frac{2w\tau_m}{(\tau_\theta + s^*\tau_p)^2} < 0$ for $w > 0$, $\tau_m > 0$. This is the information externality: a bolder time-1 use of private information makes the aggregate more informative, and at the team optimum more information always raises welfare, since the planner may ignore it.

Part (i). The linear team optimum minimizes $J(w) := J_1(w) + J_2^*(w)$ over $w \in \mathbb{R}$ (the time-2 rule being optimal given w by construction, without loss by Lemma A.3). Negative weights are dominated: J_2^* is even in w while $J_1(-w) > J_1(w)$ for $w > 0$, so attention restricts to $w \geq 0$. J is continuous with $J(w) \rightarrow \infty$ as $w \rightarrow \infty$, so a minimizer exists; $J'(0) = J_1'(0) = -2s^*/\tau_\theta < 0$ (the externality term has zero slope at $w = 0$), so every minimizer is strictly positive. Every minimizer also lies strictly below one: for $w \geq 1$,

$$\left| \frac{d}{dw} J_2^*(w) \right| = \frac{2w(s^*)^2\tau_m}{(\tau_\theta + s^*\tau_1 + s^*w^2\tau_m)^2} \leq \frac{2w(s^*)^2\tau_m}{4(\tau_\theta + s^*\tau_1)s^*w^2\tau_m} = \frac{s^*}{2w(\tau_\theta + s^*\tau_1)} < \frac{2w}{\tau_1} \leq J_1'(w),$$

the second step by the inequality $(x+y)^2 \geq 4xy$ and the third because $4w^2(\tau_\theta + s^*\tau_1) > s^*\tau_1$ for $w \geq 1$; hence $J' > 0$ on $[1, \infty)$. Let w^{**} denote any minimizer (uniqueness is not needed); at it, $J_1'(w^{**}) = -\frac{d}{dw} J_2^*(w^{**}) > 0$, which by strict convexity of J_1 gives $w^{**} > \arg \min J_1 = w(b^*)$, and $w^{**} \in (0, 1)$.

Part (ii). A perception profile assigns perceived precisions $(\hat{\tau}_\theta, p_v, p_u)$ to the prior and to the two idiosyncratic shocks v_i and u_i . By Corollary 1 only the ratios $p_v/\hat{\tau}_\theta$ and $p_u/\hat{\tau}_\theta$ move actions, so take $\hat{\tau}_\theta = \tau_\theta$ without loss of generality. By the remark below, a time- t coordination game played under perceived idiosyncratic precisions (p_v, p_u) has equilibrium equal to the conditional expectation of θ computed under the effective precisions $((1-\alpha)p_v, (1-\alpha)p_u)$: each perceived precision is further discounted by $1-\alpha$. Implementing the linear team optimum requires, first, that the time-1 weight equal w^{**} : since the time-1 rule uses only x_i , this pins the time-1 effective precision of v_i at $\hat{s}\tau_1$ with $\hat{s} := \tau_\theta w^{**}/(\tau_1(1-w^{**})) > s^*$, the inequality because $w^{**} > w(b^*)$ and the map from effective precision to weight is increasing. Second, the time-2 rule must equal $\mathbb{E}^{(s^*)}[\theta \mid x_i, m_i]$ at w^{**} . A linear rule in (x_i, m_i) determines the underlying pair of effective precisions uniquely: if the rule is $a_x x_i + a_m m_i$ with effective precisions (q_x, q_m) on $\theta + v_i$ and $\theta + u_i/w$, then $a_x = q_x/T$, $a_m = q_m/(wT)$, $T = \tau_\theta + q_x + q_m$, and (a_x, a_m) recovers $T = \tau_\theta/(1-a_x-wa_m)$ and then (q_x, q_m) . Hence matching the efficient rule forces the time-2 effective precision of v_i to equal $s^*\tau_1$. A time-constant profile gives v_i the same perceived precision at both periods, hence the same effective precision $(1-\alpha)p_v$ at both periods; but time 1 requires $(1-\alpha)p_v = \hat{s}\tau_1$ and time 2 requires $(1-\alpha)p_v = s^*\tau_1$ with $\hat{s} > s^*$, a contradiction. This covers the uniform case $p_v/\tau_1 = p_u/\tau_m = b$ a fortiori, and p_u stays free throughout.

Part (iii). Let b_1 solve $w(b_1) = w^{**}$, which exists and is unique since $w^{**} \in (0, 1)$ by part (i) and $w(\cdot)$ is an increasing bijection onto $(0, 1)$, and let the time-2 perception apply b^* to both idiosyncratic shocks. In the period-by-period distorted equilibrium defined in Section 4, time-2 agents take the actual loading w^{**} as given. Time 1 then plays w^{**} ; time 2 plays $\mathbb{E}^{((1-\alpha)b^*)}[\theta \mid x_i, m_i] = \mathbb{E}^{(s^*)}[\theta \mid x_i, m_i]$ given the information generated by w^{**} , which is the time-2 efficient rule. The allocation is the linear team optimum. Finally $b_1 > b^*$ because w is strictly increasing and $w(b_1) = w^{**} > w(b^*)$; no step used the sign of $\alpha - \alpha^*$. $\square$

Remark. The per-shock compounding used in part (ii) is Lemma A.2 with the perceived idiosyncratic covariance $\hat{\Sigma}_v$ in place of Σ_v/b : the proof goes through verbatim with $\frac{1}{b}\Sigma_v$ replaced by $\hat{\Sigma}_v$ throughout, and delivers the modified forecast in which each perceived idiosyncratic precision is further multiplied

by $1 - \alpha$. Uniform rescaling is the case $\hat{\Sigma}_v = \Sigma_v / b$ used in the body of the paper.

C.2 Proof of Corollary 3

Put $s = (1 - \alpha)b$ and $p = \tau_1 + w(b)^2\tau_m$. Under a permanent factor b , time-2 agents perceive precisions $b\tau_1$ and $b w(b)^2\tau_m$ on x_i and on $m_i/w(b)$, in the true ratio, and they take the actual loading $w(b)$ as given because the time-1 equilibrium under the shared model is $w(b)$. By the display in C.1 with $s = (1 - \alpha)b$, the time-2 rule is $k_{i2} = \gamma_p \hat{x}_i$ with $\gamma_p := sp/(\tau_\theta + sp)$ and $\hat{x}_i = \theta + \hat{v}_i$, $\text{Var}(\hat{v}_i) = 1/p$. With $\kappa^* = \theta$,

$$e_{i2} - \bar{e}_2 = \gamma_p \hat{v}_i, \quad \bar{e}_2 = (\gamma_p - 1)\theta = -\frac{\tau_\theta}{\tau_\theta + sp} \theta,$$

so

$$\text{Disp}_2 = \frac{\gamma_p^2}{p} = \frac{s^2 p}{(\tau_\theta + sp)^2}, \quad \text{Vol}_2 = \frac{\tau_\theta^2}{(\tau_\theta + sp)^2} \cdot \frac{1}{\tau_\theta} = \frac{\tau_\theta}{(\tau_\theta + sp)^2},$$

and $\text{Disp}_2 + s^* \text{Vol}_2 = Q(s, p)$ as in (16). The time-1 rule is $w(b)x_i$, the same formula with $p = \tau_1$; summing the two times gives (17). Differentiating,

$$Q_s(s, p) = \frac{2p\tau_\theta(s - s^*)}{(\tau_\theta + sp)^3}, \quad Q_p(s, p) = \frac{s[\tau_\theta(s - 2s^*) - s^2 p]}{(\tau_\theta + sp)^3}, \quad Q_p(s^*, p) = -\frac{(s^*)^2}{(\tau_\theta + s^* p)^2}.$$

By the chain rule, with

$$w'(b) = \frac{(1 - \alpha)\tau_1\tau_\theta}{(\tau_\theta + (1 - \alpha)b\tau_1)^2} > 0, \quad p'(b) = 2\tau_m w(b)w'(b) > 0,$$

the derivative is

$$\frac{dJ^c}{db} = (1 - \alpha)[Q_s(s, \tau_1) + Q_s(s, p)] + Q_p(s, p)p'(b).$$

For $s \leq s^*$ the two Q_s terms are nonpositive and $Q_p(s, p) < 0$ because $s < 2s^*$, so $dJ^c/db < 0$ on $(0, b^*]$: J^c is strictly decreasing there, and no factor at or below b^* minimizes it. If $\alpha \geq \alpha^*$ then $b^* \geq 1$, so every $b \in (1, b^*]$ has $J^c(b) < J^c(1)$. At b^* , both direct Q_s terms vanish. The remaining derivative, $Q_p(s^*, p(b^*))p'(b^*)$, gives (18) and the local improvement, since

$$p'(b^*) = 2\tau_m w(b^*)w'(b^*) > 0, \quad w'(b^*) = \frac{(1 - \alpha)\tau_1\tau_\theta}{(\tau_\theta + s^*\tau_1)^2} > 0. \quad \square$$